Journal of Geophysical Research: Solid Earth

RESEARCH ARTICLE
10.1029/2018JB015992

Key Points:
- New paleomagnetic and geochemical data from ~1144-Ma lamprophyre dikes (Ontario, Canada) show their similarity with ~1142-Ma Abitibi dikes
- A new key paleomagnetic pole that defines the North American apparent polar wander path during the late Mesoproterozoic is reported
- During ~1250- to 1110-Ma period, the plate motion rate for Laurentia was comparable with velocities of continental plates in the Phanerozoic

Supporting Information:
- Supporting Information S1

Correspondence to:
E. J. Piispa,
epiispa@yachaytech.edu.ec

Citation:
Piispa, E. J., Smirnov, A. V., Pesonen, L. J., & Mitchell, R. H. (2018). Paleomagnetism and geochemistry of ~1144-Ma lamprophyre dikes, Northwestern Ontario: Implications for the North American polar wander and plate velocities. *Journal of Geophysical Research: Solid Earth*, *123*. https://doi.org/10.1029/2018JB015992

Received 2 APR 2018
Accepted 3 JUL 2018
Accepted article online 24 JUL 2018

©2018. American Geophysical Union. All Rights Reserved.

Paleomagnetism and Geochemistry of ~1144-Ma Lamprophyre Dikes, Northwestern Ontario: Implications for the North American Polar Wander and Plate Velocities

Elisa J. Piispa[1,2], Aleksey V. Smirnov[1,3], Lauri J. Pesonen[4], and Roger H. Mitchell[5]

[1]Department of Geological and Mining Engineering and Sciences, Michigan Technological University, Houghton, MI, USA, [2]School of Geological Sciences and Engineering, Yachay Tech, Had. San Jose s/n y Proyecto Yachay, Urcuqui, Ecuador, [3]Department of Physics, Michigan Technological University, Houghton, MI, USA, [4]Department of Physics, Laboratory for Solid Earth Geophysics, University of Helsinki, Helsinki, Finland, [5]Department of Geology, Lakehead University, Thunder Bay, Ontario, Canada

Abstract We present new paleomagnetic and geochemical data from a suite of the ~1144-Ma ultramafic lamprophyre dikes that outcrop in the Canadian Shield northeast of Lake Superior (Ontario, Canada). Nineteen of 22 sampled dikes yielded consistent characteristic remanent magnetization directions of normal (n = 5) and reversed (n = 14) polarity. The primary origin of characteristic remanent magnetization is bolstered by positive baked contact tests and a reversal test. The group mean direction (D = 306.4°, I = 72.1°, α_{95} = 5.5°, N = 19) obtained from the lamprophyre dikes is statistically indistinguishable from the group mean direction (D = 297.4°, I = 65.5°, α_{95} = 8.3°, N = 8) previously reported for the nearly coeval ~1142-Ma Abitibi dikes. The geochemistry of the lamprophyre dikes suggests strong affinity with magmas derived from ocean island basalt-type mantle sources, consistent with the mantle plume hypothesis for the formation of the ~1.1-Ga North American Midcontinent Rift. The similarity in age, trend, paleomagnetism, and geochemistry indicates that the lamprophyre and Abitibi dike suites represent the earliest magmatic event associated with the commencement of rifting. The combined mean direction (D = 303.1°, I = 70.2°, α_{95} = 4.5°, N = 27) corresponds to a paleomagnetic pole at P_{lat} = 55.8°N, P_{long} = 220.0°E (A_{95} = 7.3°). The new pole merits the highest classification on the Q-scale of paleomagnetic reliability and represents a key pole defining the North American apparent polar wander path during the late Mesoproterozoic. Combined with high-quality data from the ~1108-Ma Coldwell Complex, our data indicate an equatorward motion of Laurentia at 3.8 ± 1.4 cm/year, comparable with the present-day velocities of continental plates, before switching to extremely rapid motion between ~1108 and ~1099 Ma.

Plain Language Summary Similar to a magnetic tape, rocks can retain the direction of ancient Earth's magnetic field. Scientists use this record (known as paleomagnetism) to reconstruct past positions of continents and to decipher the geological history of our planet. We investigated paleomagnetism and chemical composition of the ~1.14 Ga-old intrusive rocks called lamprophyres exposed in Northwestern Ontario (Canada). We found that the paleomagnetic field directions recorded in lamprophyres are indistinguishable from those recorded by another similar age suite of basaltic intrusions called the Abitibi dikes, from the same area. The combined data from these rocks allowed us to constrain the position of an ancient supercontinent called Laurentia at ~1.14 billions of years ago more accurately than it was possible before. Our results convincingly show that, during that time, Laurentia moved with a velocity comparable to present-day plate velocities, before switching to an extremely rapid motion approximately 35 millions of years later. The lamprophyre and Abitibi rocks also share similar chemical signatures, close to those observed for ocean island basalts (e.g., Hawaii). These observations support the hypothesis that a failed ocean opening attempt called the North American Midcontinent Rift was instigated by the arrival of a hot mantle material upwelling to the Earth surface.

1. Introduction

The ~1.1-Ga North American Midcontinent Rift (MCR), which ranges from Kansas through Lake Superior and Lower Michigan (Figure 1), represents the best preserved example of a Proterozoic intracontinental rift (Van Schmus & Hinze, 1985). The rift magmatism resulted in eruption of >300,000 km^3 of predominantly mafic,

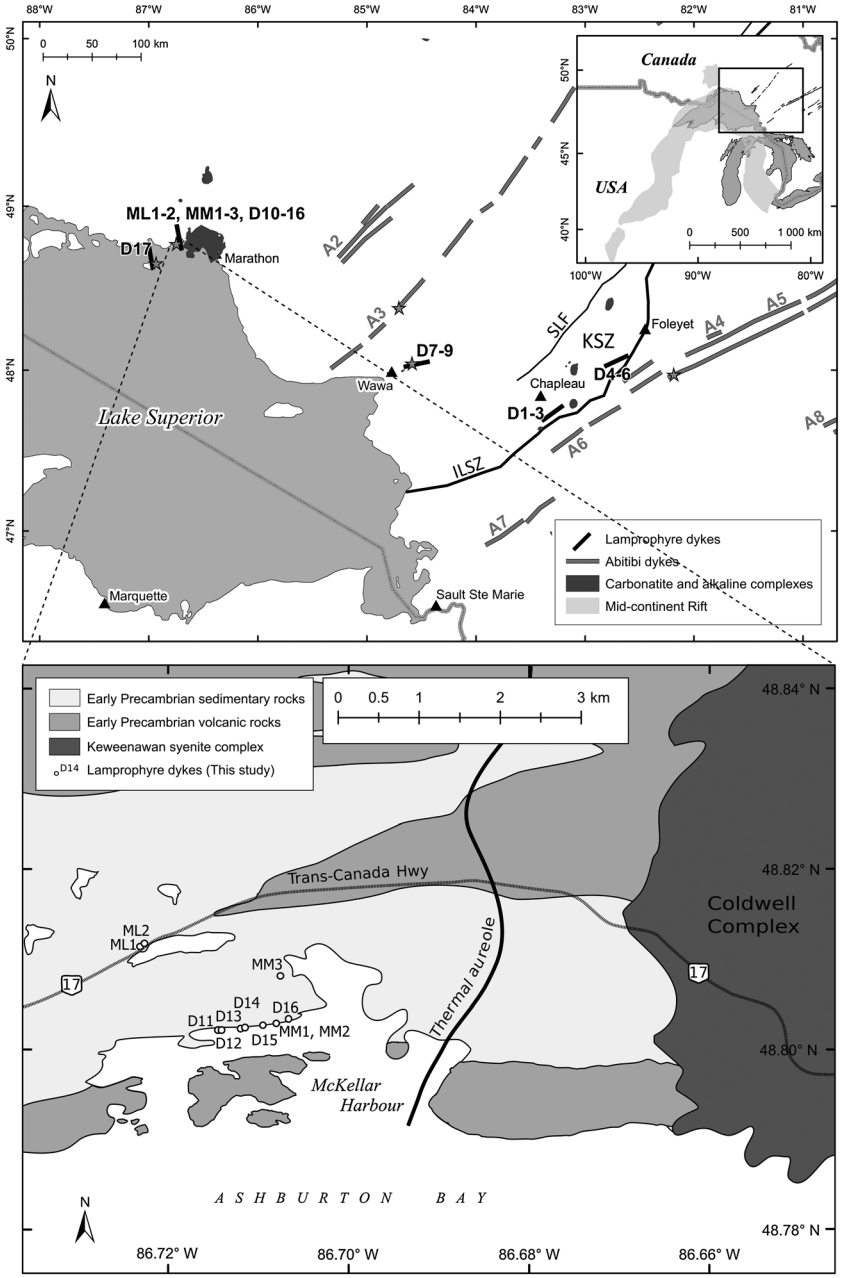


Figure 1. Locations of sampling sites of the lamprophyre dikes (black lines show approximate dike trends; see Table 3 for the site coordinates). Gray lines and labels A2–A8 show the approximate locations of the Abitibi dike sites investigated by Ernst and Buchan (1993). Stars indicate the approximate sites used for U-Pb isotope geochronology (Table 1). The dark gray areas in the upper plot show the carbonatite and alkaline complexes related to the Midcontinent Rift (MCR). Kapuskasing Structural Zone (KSZ) is bounded by the Ivanhoe Lake Shear Zone (ILSZ) and Saganash Lake Fault (SLF). The inset shows the position of our sampling area (black outlined box) with respect to the MCR system (light gray shaded area). The lower plot shows the sampling site locations in the East Lake Superior area with respect to the Coldwell Complex and its thermal aureole.

mantle-derived magma and emplacement of possibly an equal amount of intrusive rocks (Davis & Green, 1997; Green et al., 1987; Hutchinson et al., 1990). Based on high precision U-Pb geochronology, the MCR magmatism has been divided into a ~1145- to 1140-Ma precursor stage, two main pulses at ~1115–1105 and ~1100–1094 Ma, and a period of waning volcanic activity at circa 1087–1085 Ma (e.g., Davis & Green, 1997; Heaman et al., 2007).

Table 1
U-Pb Isotopic Ages of the Lamprophyre and Abitibi Dikes

Dike/unit	Sample ID	Lat (°N)	Long (°W)	Age (Ma)	Method	Ref
East Lake Superior lamprophyre	LH87-10 (D9)	48.04	84.54	1143 ± 12	U-Pb perovskite	1
Marathon lamprophyre	LH85-4B (D15)	48.80	86.71	1145 + 15/−10	U-Pb perovskite	1
Slate Island lamprophyre	LH91-8 (D17)	48.63	87.03	1141 ± 9	U-Pb perovskite	1
Kapuskasing Uplift lamprophyres	KL-1, KL-2, LL1	N/A	N/A	1144 ± 7	$^{40}Ar/^{39}Ar$ phlogopite	1
Wawa Abitibi dike	(A3)	48.36	74.73	1142 ± 3	U-Pb baddeleyite	2
Great Abitibi dike	(A6)	47.95	82.23	1141 ± 2	U-Pb baddeleyite	3

Note. Sample ID refers to the name used in the geochronology study, or, in parentheses, the site name studied for paleomagnetism in this study (lamprophyres) and in Ernst and Buchan (1993; Abitibi dikes). Lat and Long are the approximate site latitude and longitude; Ref stands for the age reference: 1, Queen et al. (1996); 2, L. M. Heaman, unpublished data cited in Sage (2000), and 3, Krogh et al. (1987).

Restoring Mesoproterozoic plate motions through paleomagnetic research is of key importance to constrain deep-time geodynamics; however well-dated rocks of that age carrying the primary paleomagnetic signal are sparse. Paleomagnetic, geochronological, and geochemical analyses of the MCR rocks have proved useful for deciphering the evolution and geodynamical context of the rift and for constructing the late Mesoproterozoic apparent polar wander path (APWP) for Laurentia (Halls & Pesonen, 1982; Hnat et al., 2006; Hollings et al., 2010; Kulakov et al., 2014; Palmer & Davis, 1987; Swanson-Hysell et al., 2009). However, while the main and final stages of MCR have been represented by several high-quality paleomagnetic data sets (Fairchild et al., 2017; Kulakov et al., 2013; Kulakov et al., 2014; Swanson-Hysell et al., 2009, 2014; Tauxe & Kodama, 2009), the paleomagnetic data for the precursor stage remain limited.

The ~1144-Ma ultramafic lamprophyre dikes (Queen et al., 1996) and the ~1142 Abitibi diabase dikes (Krogh et al., 1987) that crop-out northeast of Lake Superior (Ontario, Canada; Figure 1) represent the earliest magmatic events related to MCR (Table 1). For both suites, local dike trends form a fan pattern radiating from a common center in the Lake Superior basin (Figure 1). Based on similar age and radiating dike trends, Queen et al. (1996) suggested that the lamprophyre and Abitibi dikes represented the incipient magmatic activity of the Midcontinent rifting, preceding the main rift magmatism by ~35 Ma.

In a previous paleomagnetic investigation of the Abitibi dikes, Ernst and Buchan (1993) reported a group mean pole based on five normally (N) magnetized and two reversely (R) magnetized dikes; two independent N-polarity directions were obtained from distinct lithological units within the Great Abitibi Dike (site A6, Figure 1). The primary nature of the N-polarity directions in the Abitibi dikes has been established by a baked contact test against a Matachewan dike (Ernst & Buchan, 1993). The absence of pervasive regional remagnetization was also inferred from the presence of opposite polarity directions in the neighboring dikes. Although both R-polarity directions manifested inclinations steeper than their N-polarity counterparts, the authors interpreted them to be primary and symmetrical to the N-directions based on the similarity of group mean poles calculated from the N-polarity data and from the combined polarity data. Nevertheless, the primary origin of the R-polarity remanence in the Abitibi dikes remains tentative.

Later, one of the N-polarity dikes (site A1) studied by Ernst and Buchan (1993) was shown to belong to the ~2167-Ma Biscotasing swarm (Halls et al., 2005). Halls et al. (2005) also reported a new N-polarity direction from another Abitibi dike. The modified group mean direction for the Abitibi dikes (D = 297.4°, I = 65.5°, α_{95} = 8.3°, N = 8) corresponds to a mean pole at P_{lat} = 50.5°N, P_{long} = 213.8°E (A_{95} = 12.5°) offset by 8.5° from the original pole. Nevertheless, the original Abitibi pole has been routinely utilized as a key pole defining the Laurentian APWP in the late Mesoproterozoic (e.g., Pisarevsky et al., 2014). However, being derived from a limited number of independently cooled units, the pole does not represent the time-averaged field, and hence, the reconstructions based on this pole are likely to be biased.

Paleomagnetism of the ~1144-Ma lamprophyre dikes has not been previously investigated. Here we report the results of a detailed paleomagnetic, rock magnetic, and geochemical investigation of the lamprophyre dikes conducted in order to test their hypothesized connection with the ~1142-Ma Abitibi dikes and to refine the Laurentian APWP for the late Mesoproterozoic. We also discuss the implications of our new data for the plate velocities of Laurentia during the Late Mesoproterozoic as well as for the origin and tectonic settings of the MCR.

2. Geological Setting

The study area is located in the southern Superior Province of the Canadian Shield consisting of volcanic-plutonic and metasedimentary-gneissic subprovinces (Figure 1). Northeast of Lake Superior, the east-west trending Wawa and Abitibi metavolcanic greenstone subprovinces are transected by the northeast trending Kapuskasing Structural Zone and are intruded by multiple Proterozoic dike swarms including the Abitibi dikes (e.g., Buchan & Ernst, 2004; Figure 1). In addition, several MCR-related ~1105- to 1110-Ma carbonatite and alkaline complexes are exposed in proximity to the Abitibi and lamprophyre dikes (e.g., Costanzo-Alvarez et al., 1993; Ernst & Bell, 2010; Sage, 1991a; Stott & Josey, 2009; Figure 1).

In this study, we divide the lamprophyre dikes based on their location into the Eastern Lake Superior (ELS; sites D1-9) and Marathon areas (sites ML1-2, MM1-3 and D10-17; Figure 1). The Marathon area lamprophyres intrude the metavolcanic-metasedimentary sequence of Archean Schreiber-White River greenstone belt but do not crosscut the adjacent ~1108-Ma alkaline Coldwell Complex (Platt et al., 1983). On Slate Island, a lamprophyre dike (site D17) is cut by a ~1100-Ma Keweenawan mafic dike (Sage, 1991b). In the ELS area, the lamprophyre dikes crosscut the Archean rocks as well as the Paleoproterozoic Matachewan, Senneterre, Biscotasing, Marathon, Fort Frances, and Kapuskasing dikes but do not intersect the ~1100-Ma alkaline and carbonatite complexes (Queen et al., 1996). No crosscutting relationship between the lamprophyre and Abitibi dikes has been observed.

The lamprophyre dikes are narrow (<1 m), fine- to medium-grained and show dark gray-green color on fresh surfaces and brownish color on weathered surfaces. They dip mainly vertically and their strikes fan from a locus in Lake Superior (Figure 1). Many dikes exhibit sharp contacts with the country rock and occasionally have chilled margins. The whole rock and mineral compositions of the lamprophyre dikes in the Marathon area were previously investigated by Platt and Mitchell (1982) and Platt et al. (1983). The ultramafic lamprophyres consist of olivine and titanium phlogopite phenocrysts within a groundmass consisting of a mixture of titanian phlogopite, primary carbonate, spinel, apatite, perovskite**,** and melilite. The spinel compositions range from titanian-aluminum-chromites to ulvospinel-magnetites and indicate affinities with alkaline rock-carbonatite magmatism.

3. Results

3.1. Sampling

We collected 158 independently oriented samples from 22 lamprophyre dikes from the Marathon and ELS areas (Figure 1). At two sites (D1 and MM2), samples were also obtained from the baked and unbaked host rocks for baked contact tests. At 19 sites, oriented cores were taken using a gasoline-powered hand drill and, at three sites (D17, MM1, and MM2), oriented block samples were collected. Orientation was recorded with both solar and magnetic compasses. Samples for geochemical analyses were collected at 17 sites in the Marathon and ELS areas. No dikes sampled in the ELS area exhibited mild deformation or shearing described in Queen et al. (1996).

3.2. Geochemistry

Whole-rock geochemical analyses were performed by Activation Labs Ltd (Ancaster, Canada; Actlabs package 4Lithores) with lithium metaborate/tetraborate fusion using a combination of simultaneous or sequential Thermo Jarrell-Ash ENVIRO II Inductively Coupled Plasma (ICP) or a Varian Vista 735 ICP for the major oxides and Perkin Elmer Sciex ELAN 6000, 6100, or 9000 ICP/MS for the trace elements. The whole rock analyses of the major and trace elements were performed on six dikes from the ELS area and 11 dikes from the Marathon area (Table 2). One of these dikes (D17) was noticeably weathered and has the lowest total iron, K_2O, and MgO contents and highest CaO content and loss-on-ignition (LOI) values (Table 2). The trace element data from this dike fall within the variation of the other lamprophyres except for higher amounts of Hf and Pb (Table 2). However, because of the substantial weathering, we excluded the geochemical data from D17 from the following discussion.

The lamprophyre dikes are characterized by low SiO_2 content (22–35 wt %), moderate to high total iron (11–19 wt %), high CaO (11–22 wt %), TiO_2 (2.56–5.18 wt %), and MgO (average ~16 wt %) and low Al_2O_3 (average ~4 wt %) and Na_2O (0.05–2.46 wt %; Table 2). The major element compositions of the ELS

Journal of Geophysical Research: Solid Earth
10.1029/2018JB015992

Table 2
Major and Trace Element Compositions of the Late Mesoproterozoic Lamprophyre Dikes Around Lake Superior, Canada

		East Lake Superior area lamprophyres						Marathon area lamprophyres										
Chemical element	DL	D1	D2	D4	D5	D7	D9	ML2	D10C	D10M	D11	D12	D13-1	D13-2	D14	D15-M	D16	D17
Major Oxides (wt %)																		
SiO_2	0.01	27.38	30.30	35.36	25.77	28.81	25.19	28.78	25.46	26.63	21.59	32.59	29.88	28.12	28.59	31.88	24.93	25.69
TiO_2	0.001	3.03	2.87	2.75	2.56	4.58	3.90	3.19	2.87	3.20	2.65	3.69	4.10	4.00	5.18	4.25	3.39	3.47
Al_2O_3	0.01	4.56	4.10	9.74	3.70	3.82	3.17	3.03	3.54	3.77	3.01	4.69	3.48	3.37	3.56	4.04	3.13	3.56
$Fe_2O_3{}^T$	0.01	12.31	13.42	13.76	11.15	16.98	13.61	10.87	14.69	14.93	11.94	11.75	12.50	12.63	18.19	18.53	12.27	6.40
MnO	0.001	0.20	0.20	0.25	0.19	0.24	0.27	0.26	0.32	0.26	0.24	0.25	0.23	0.23	0.27	0.25	0.30	0.27
MgO	0.01	12.26	14.72	8.59	18.55	17.28	12.86	18.46	15.95	17.12	13.58	16.38	14.44	14.33	17.68	20.30	17.19	6.38
CaO	0.01	14.15	12.44	15.22	11.36	12.91	14.21	12.67	19.61	17.67	22.06	13.56	16.82	17.44	11.94	11.46	16.24	27.75
Na_2O	0.01	2.15	2.07	2.46	1.98	0.43	2.09	0.15	0.16	0.05	0.07	0.11	0.12	0.14	0.07	0.09	0.16	0.11
K_2O	0.01	2.06	1.29	1.47	2.37	2.13	1.91	1.77	1.70	1.38	1.51	1.81	1.47	1.38	1.52	1.63	1.33	0.89
P_2O_5	0.01	0.17	0.11	0.83	0.84	1.14	1.11	0.48	3.17	2.89	2.79	1.02	0.82	0.79	0.89	0.84	1.27	1.04
LOI		20.62	16.89	8.89	19.45	9.89	22.43	18.64	10.86	10.16	18.07	13.95	14.89	15.85	12.68	7.35	19.87	24.87
Total	0.01	98.87	98.40	99.32	97.92	98.21	100.70	98.29	98.32	98.05	97.53	99.79	98.76	98.27	100.60	100.60	100.10	100.40
Trace elements (ppm)																		
Be	1	7	2	4	4	2	4	3	3	3	3	3	2	2	4	3	3	3
Sc	1	18	21	28	12	22	18	18	24	22	18	24	21	20	21	27	19	28
V	5	219	234	303	172	321	272	187	260	212	225	287	293	276	336	387	235	379
Cr	20	410	530	180	990	540	450	1000	550	640	480	730	730	720	660	730	520	640
Co	1	67	80	54	71	89	78	83	64	69	53	79	82	79	87	87	71	70
Ni	20	390	600	160	650	560	480	700	420	490	360	510	510	520	500	550	460	410
Cu	10	80	100	110	80	130	120	80	90	120	90	120	100	90	80	150	80	160
Zn	30	150	100	130	170	130	120	100	110	140	120	130	110	100	130	120	120	140
Ga	1	18	14	21	12	20	19	14	15	16	13	18	17	16	20	18	15	17
Ge	0.5	1.4	1.4	1.6	1.1	1.7	1.3	1.4	2.9	2.1	2.2	2.7	3.3	3.5	2.9	2.9	2.6	1.8
As	5	-	-	-	9	-	-	-	39	28	51	54	53	56	31	31	21	7
Rb	1	66	53	51	98	80	88	74	71	54	62	85	54	49	57	86	60	44
Sr	2	1096	480	1021	1304	788	1133	1153	1314	1306	1895	1044	788	730	1001	519	1456	675
Y	0.5	47.3	21.7	34	25.8	30.3	33	19.6	52.6	44.6	40.4	28.4	23.6	22.6	27.5	22.1	30.7	40.8
Zr	1	265	217	291	201	405	339	325	676	552	539	309	369	357	367	622	342	719
Nb	0.2	115	56.9	103	103	97.2	86.6	129	230	218	146	130	115	114	139	145	217	179
Mo	2	24	-	2	-	-	3	2	-	-	-	6	4	4	2	2	3	-
Ag	0.5	1.2	0.9	1.2	0.9	1.7	1.5	1.4	4.7	3	3.3	1.9	2.2	2.1	2.1	3.5	2	3.2
In	0.1	0.1	-	-	-	0.1	-	-	-	-	-	-	-	-	-	-	-	0.2
Sn	1	3	3	2	2	3	3	2	3	-	-	-	3	3	3	4	3	4
Sb	0.2	1.4	0.7	-	0.3	-	-	-	0.8	0.6	2	1.1	2.3	2.1	0.8	0.8	2	0.8
Cs	0.1	1.0	1.0	0.8	2.3	1.7	2.8	2.1	1.2	1.1	2.5	8.1	2.8	2.4	2.9	3.3	1.9	1.1
Ba	3	1001	319	1231	1137	782	567	1150	1509	1245	1168	873	710	690	690	909	1073	841
La	0.05	108	51.6	97	110	82.3	80.3	113	201	223	188	123	101	100	129	83	185	116
Ce	0.05	230	109	179	215	182	172	229	461	526	432	259	221	223	291	165	405	224
Pr	0.01	29.3	13.5	20.7	26.5	23.1	22	27.4	57.1	67.1	54.2	31.6	27	27.6	35.5	19.5	47.6	26.6
Nd	0.05	121	54	80.6	101	94.4	86.6	105	224	266	213	125	108	109	141	73.4	179	104
Sm	0.01	21.6	9.5	14.0	16.9	16.8	15.3	16.5	36.7	42.9	35.1	20.5	18.4	18.2	24.2	13.0	28.3	16.0
Eu	0.005	6.34	2.95	4.1	4.74	4.9	3.95	4.39	9.77	11.2	9.26	5.29	4.65	4.55	6.44	3.42	6.69	5.34
Gd	0.01	16.8	7.66	10.1	10.9	12.5	9.84	10.1	22.6	25.5	21.6	13.7	11.4	11.2	14.6	8.17	16.2	11.5
Tb	0.01	2.5	1.08	1.36	1.35	1.66	1.24	1.16	2.73	3.03	2.47	1.68	1.33	1.29	1.72	1.05	1.79	1.55
Dy	0.01	12	4.86	6.72	5.8	7.25	6.11	4.99	12.1	12.3	10.6	6.98	5.54	5.55	6.98	4.82	7.53	7.62
Ho	0.01	1.68	0.8	1.14	0.9	1.14	1.1	0.71	1.84	1.79	1.57	1.11	0.88	0.83	1.07	0.77	1.13	1.28
Er	0.01	3.69	1.9	2.82	2.06	2.51	2.92	1.63	4.34	4.08	3.69	2.46	1.89	1.85	2.32	1.74	2.57	3.16

Table 2 *(continued)*

Chemical element	DL	East Lake Superior area lamprophyres						Marathon area lamprophyres										
		D1	D2	D4	D5	D7	D9	ML2	D10C	D10M	D11	D12	D13-1	D13-2	D14	D15-M	D16	D17
Tm	0.005	0.421	0.261	0.352	0.236	0.277	0.419	0.176	0.497	0.442	0.425	0.283	0.213	0.202	0.249	0.205	0.272	0.395
Yb	0.01	2.58	1.47	2.2	1.15	1.43	2.62	0.84	2.57	2.06	2.01	1.49	1.04	1.05	1.15	1.08	1.45	2.39
Lu	0.002	0.36	0.189	0.293	0.14	0.173	0.346	0.105	0.3	0.259	0.24	0.201	0.129	0.133	0.143	0.139	0.176	0.32
Hf	0.1	6.7	5.6	5.9	4.3	9.2	7.8	6.6	8.7	6.5	7.6	6.6	6.8	6.5	6.9	11.9	6.5	14.3
Ta	0.01	6.4	4.32	5.11	6.72	7.26	6.61	8.88	9.73	15.3	9.71	8.72	8.31	8.16	10.2	8.84	12.5	7.88
W	0.5	10.5	3.5	1	1.3	-	2.7	1.7	1.8	1.8	1.1	2.6	3.9	4.5	2.8	2.4	4.1	2.4
Tl	0.05	0.74	0.22	0.11	0.28	0.47	0.42	0.54	-	1.53	0.68	1.1	-	-	-	-	-	0.27
Pb	5	11	-	8	8	-	-	7	15	10	8	11	7	6	9	9	20	34
Bi	0.1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Th	0.05	43.9	9.21	10.3	15.5	7.12	12	16.7	34.2	41.5	33.1	12.9	11.2	12.6	14.2	9.9	26.2	10.5
U	0.01	2.13	1.07	2.81	2.91	2.94	3.39	3.33	10.2	8.36	8.43	4.61	2.51	2.47	3.09	3.14	5.56	3.56

Note. DL = detection limit; $Fe_2O_3^T$ = total iron as ferric iron; LOI = loss on ignition; - = below detection limit.

and Marathon area dikes (this work; Platt & Mitchell, 1982) are similar except for a higher Na_2O content in the ELS samples. The dikes are classified as ultramafic lamprophyres or aillikites using the CaO–MgO–Al_2O_3 and Fe_2O_3(t)–MgO–Al_2O_3 ternary diagrams, respectively (Rock, 1986, 1987).

All of the analytical results for the lamprophyre dikes show very high LOI values (7–22 wt %), which are consistent with those reported by Platt and Mitchell (1982) for the Marathon lamprophyres. The high LOI values reflect the combined effect of the primary high volatile content (CO_2 and H_2O) and of the low-temperature alteration and weathering. Consequently, the major element compositions and mobile trace elements such as Cs, Rb, Ba, K, and Sr must be interpreted with caution, since they are susceptible to mobility through low-grade alteration and weathering. For this reason, much of our geochemical interpretation is based on the data on relatively immobile high field strength trace elements (e.g., Zr, Hf, and Nb), following Pearce (2008).

The immobile trace elements (e.g., rare earth element [REE], Zr, Hf, Nb, Ta, and Y) provide better insight into the variations among the lamprophyre dikes and for comparison with the Abitibi dikes. The REE plots are coherent (Figure 2a) and therefore reflect primary magmatic values. All the lamprophyres are enriched in light REE relative to heavy REE and, hence, have steeper REE distribution patterns than alkali basalts (e.g., compare the lamprophyre average $(Ce/Yb)_N \approx 48$ from this study to average oceanic island basalts (OIBs; $(Ce/Yb)_N \approx 10$ from Sun & McDonough, 1989). Overall, the trace element patterns normalized to chondrites are similar to those classified as ultramafic lamprophyres by Rock (1987).

Positive correlations in binary trace element diagrams, such as Th/Yb versus Ce/Yb, Hf versus Zr and Nb versus Zr (Figures 2a–2c), suggest broadly similar sources for the ELS and Marathon lamprophyres. In comparison with the Marathon lamprophyres, the ELS lamprophyres are less enriched in light REE (Figures 2b–2d) and fall in the lower range for all immobile trace elements (Figures 2b–2d). This is consistent with lower degrees of partial melting for the Marathon lamprophyres with respect to the ELS lamprophyres. The Th-Nb proxy for crustal input (Pearce, 2008) shows that the lamprophyres have an intraplate, OIB affinity (Figure 2e).

3.3. Rock Magnetic Properties

The rock magnetic properties of the lamprophyre dikes were investigated at Michigan Technological University. We measured magnetic hysteresis parameters (H_c, coercivity; H_{cr}, coercivity of remanence; M_s, saturation magnetization; and M_{rs}, saturation remanence) using a model 2900 Princeton Measurement Corporation alternating gradient magnetometer. Whereas the majority (24 of 31) of the magnetic hysteresis loops exhibit a regular shape, seven samples revealed "pot-bellied" loops indicating a mixture of single domain and superparamagnetic particles (e.g., Tauxe et al., 1996; Figures S1a and S1b in the supporting information). The M_{rs}/M_s and H_{cr}/H_c ratios measured from the Marathon area dikes are well grouped plotting within the pseudo single domain region of the Day plot (Day et al., 1977; Dunlop, 2002), whereas the ELS area dikes show a greater variation from nearly single domain to nearly multidomain behavior (Figure S1c).

We measured the dependence of low-field magnetic susceptibility (κ) on temperature (T) as the samples were cycled in argon from ambient temperature to 700 °C with a model MFK-1FA AGICO magnetic susceptibility meter. The high-temperature thermomagnetic runs were preceded and followed by measurement of the $\kappa(T)$ curves between −192 °C and ambient temperature in air. Most measured curves showed the presence of a single magnetic phase with a Curie temperature (T_c) between 570 and 585 °C (Figures S1d–S1i). A susceptibility peak at ~153 °C, characteristic of the Verwey transition (Verwey, 1939), indicates that this phase is nearly stoichiometric magnetite (Figure S1). In some instances, $\kappa(T)$ curves also indicated a phase with $T_c \approx 320$ °C, most likely, pyrrhotite

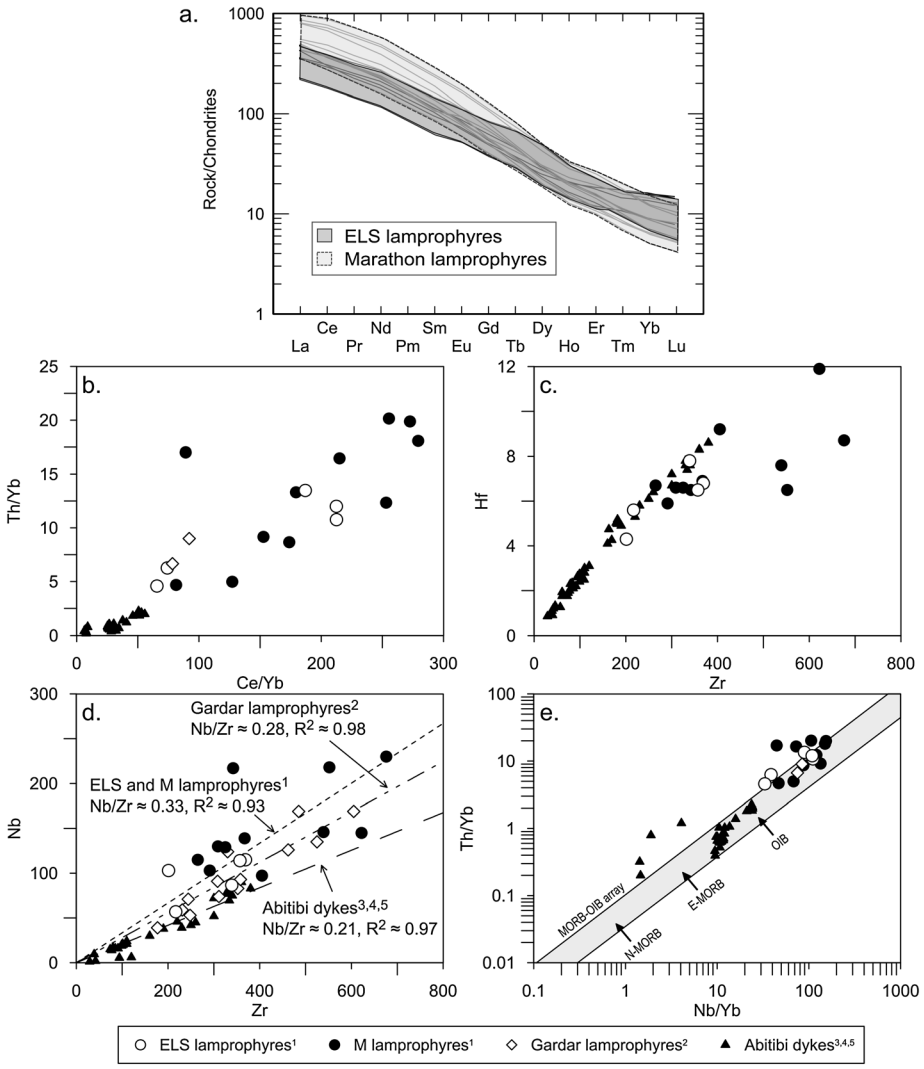


Figure 2. (a) Rare earth element profiles of the East Lake Superior (ELS) and Marathon (M) area lamprophyres. Samples were normalized to the chondrites after Sun and McDonough (1989). (b–e) Binary trace element diagrams of ELS and M lamprophyres. Data from the Gardar lamprophyres and Abitibi dikes are also shown when available. (b) Th/Yb versus Ce/Yb, (c) Hf versus Zr, (d) Nb versus Zr, and (e) Th/Yb versus Nb/Yb after Pearce (2008). References in the legends: (1) This study: ELS and M lamprophyres; (2) Goodenough et al. (2002): Gardar lamprophyres; and (3) Ernst & Buchan, 2010; (4) Halls et al. (2005) and (5) Halama et al. (2007): Abitibi dikes. OIB = oceanic island basalt; MORB = mid-ocean ridge basalt.

(Figure S1e). No systematic difference in thermomagnetic behavior between the ELS and Marathon area dikes was observed. Both nearly reversible and irreversible $\kappa(T)$ curves were measured indicating variable degree of heating-induced alteration, sometimes even within the same dike (e.g., Figures S1d and S1e).

The opaque mineralogy of several representative samples was examined using an Olympus PMG3 inverted research metallurgical reflected light microscope at Michigan Technological University and a Hitachi SU-70 Schottky Field Emission Scanning Electron Microscope at Lakehead University. Both techniques revealed complex magnetomineralogical compositions of the lamprophyres represented by various iron oxides and/or sulfides (Figure S2). The reflected light images of the opaque minerals show a range from strongly/moderately altered to relatively unaltered grains (Figures S2a–S2d). Almost all samples show the effects of both deuteric and secondary alteration. Deuteric oxidation up to the R7 stage for titanohematite (Haggerty, 1991) was observed in some samples indicated by completely pseudomorphed primary ilmenite crystals with pseudobrookite, rutile, and hematite (Figure S2a). Some dikes, however, contained better-

preserved iron oxide grains with lower oxidation stages (Figures S2b–S2d). These grains typically have relict titanomagnetite or chromian spinel cores with secondary titanomaghemite oxidation rims around the grain boundaries (Figures S2b and S2c) or fairly well-preserved titanomagnetite grains containing sandwich or trellis-type lamellae (Haggerty, 1991; Figure S2d).

All analyzed samples contained varying (generally, small) amounts of pyrite/pyrrhotite (Figures S2a and S2d; Platt & Mitchell, 1982). In addition to the low-temperature Ti-maghemite oxidation rims, the effects of weathering were observed in some instances in the form of brown granular matrix coating the primary and secondary minerals possibly consisting of a hydrated mixture of clay minerals, oxides, and hydroxides including limonite and goethite. In general, the samples that contained strong to moderately altered and/or weathered primary titanomagnetite and/or ilmenite grains corresponded to the dikes that yielded uninterpretable, or lower quality, paleomagnetic results.

Backscattered electron images of the oxide phases show Ti-magnetite and ilmenite grains with different degrees of alteration as well as some relict chromian spinels (Figures S2e–S2h). Some dikes from the ELS area contain anatase (TiO_2) in association with REE carbonates and ilmenite (e.g., Figure S3h). These dikes are typically distinct from the other dikes from the ELS area by showing the highest increase in low-field susceptibility during the thermomagnetic analysis (e.g., Figure S1i).

Overall, rock magnetic analyses and interpretations are consistent with the results of electron microscopy and indicate that the primary remanent magnetization has been preserved by the dikes.

3.4. Paleomagnetism

We measured remanent magnetization of our samples with a 760-R superconducting rock magnetometer (2G Enterprises) integrated with an in-line alternating field demagnetizer. Following the natural remanent magnetization measurement, the samples were subjected to low-temperature demagnetization by submerging them into liquid nitrogen for approximately 1–1.5 hr (Schmidt, 1993). Next, sister specimens from each sample were subjected to detailed thermal and alternating field demagnetization. Thermal demagnetization was conducted in a nitrogen atmosphere with a model TD-48SC ASC thermal demagnetizer. The progressive demagnetization was performed in 15–20 steps until the measured magnetization directions became unstable or until the remanence magnitude decreased below noise level.

Natural remanent magnetization of all measured samples was composed of one or two components (Figure 3). When present, the magnetically soft component was destroyed by 20–25 mT or by 400–450 °C (e.g., Figure 3f). The directions of secondary component were scattered but, in general, had northerly declinations with intermediate downward inclinations possibly indicating a recent field overprint (e.g., Figure 3f).

The characteristic remanent magnetization (ChRM) component was calculated for samples that displayed nearly linear demagnetization trajectories toward the origin of a vector plot. The best line fitting was used if the trajectory was defined by at least four consecutive demagnetization steps with a maximum angular deviation $<10°$. However, for some samples, the remanence components had overlapping unblocking spectra resulting in remagnetization circles on the stereonet (Figures 3e–3g). To calculate the site mean directions for these dikes, we used either the least squares method for finding the remanence direction from converging great circles or a combined approach in which the remanence directions defined by the best line fitting for certain specimen(s) are used as an anchor for the remagnetization circles from the other specimens (McFadden & McElhinny, 1988).

The Fisher statistics (Fisher, 1953) was used to calculate the site and group mean directions. The site mean directions were accepted for further analysis if their 95% confidence circle (α_{95}) was less than 15°. We further divided the accepted directions in two classes. Class A includes the directions calculated from four or more samples ($N > 3$) and exclusively based on best line fitting. Class B includes directions partly or completely based on remagnetization circle fitting and/or calculated from two or three samples per dike. Nine Class A and ten Class B site mean directions of both normal (N) or reversed (R) polarity were obtained (Table 3). The normal/reversed polarity corresponds to downward/upward inclination (e.g., Robertson & Fahrig, 1971). The data from three remaining dikes were rejected because of a significant scatter of the ChRM directions.

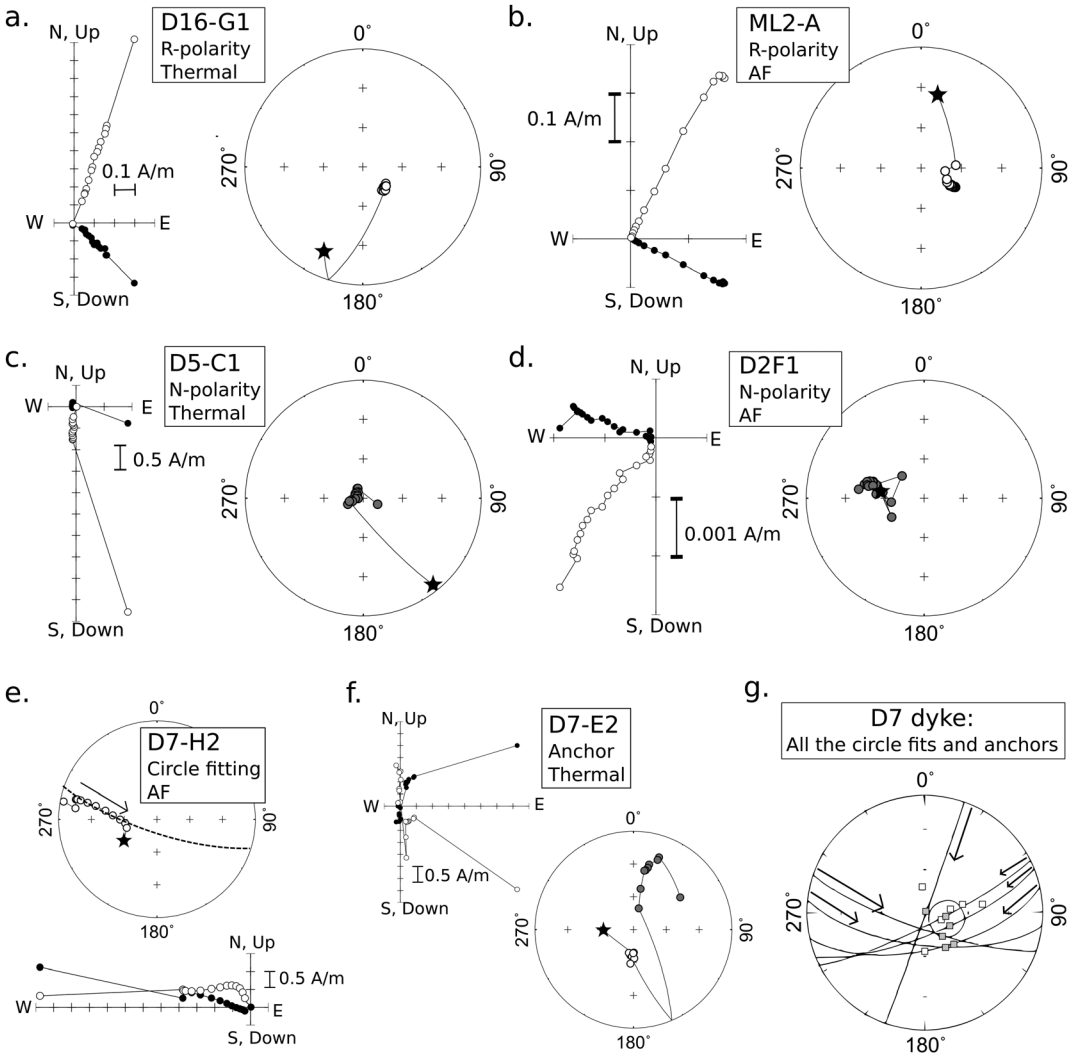


Figure 3. Examples of the demagnetization behavior of the lamprophyre dikes. (a–f) Vector endpoint diagrams (open/closed symbols show the endpoint projections on vertical/horizontal plane and division units are marked next to the plot) and equal area plots (open/closed symbols refer to reversed/normal polarity and the star shows the direction after the last demagnetization step), R (N) polarity refers to reversed (normal) polarity and thermal (AF) indicates thermal (alternating field) demagnetization. (a, b) Reverse polarity dikes (D16 and ML2) from Marathon area, (c, d) normal polarity dikes D5 and D2 from East Lake Superior area, and (e–g) examples of a circle fitting, anchor, and all the data combined in one plot for dike D7.

Ten dikes from the Marathon area and four dikes from the ELS area yielded steep R-polarity ChRM directions (Table 3 and Figure 4a). The group mean directions based on Class A ($D = 116.4°$, $I = -66.1°$, $\alpha_{95} = 16.1°$, $N = 4$) and Class B ($D = 131.5°$, $I = -71.2°$, $\alpha_{95} = 7.3°$, $N = 10$) data are statistically indistinguishable resulting in the combined mean direction of $D = 126.5°$, $I = -69.9°$ ($\alpha_{95} = 6.3°$, $N = 14$; Figures 4a and 4e). In addition, three dikes from the ELS area and two dikes from the Marathon area yielded Class A directions of normal polarity (the mean direction: $D = 305.9°$, $I = 78.3°$, $\alpha_{95} = 13.7°$; Table 3 and Figures 4a and 4e).

To assess the reliability of the paleomagnetic directions, we conducted two baked contact tests. One positive test was obtained from a 30-cm-wide dike (D1) cutting through a ~10-m-wide Matachewan dike in the ELS area. Two baked Matachewan samples collected within 5–7 cm from the contact yielded steep R-polarity directions ($D = 60.9°$, $I = -75.6°$, $\alpha_{95} = 7.9°$) similar to the site mean direction ($D = 182.7°$, $I = -78.1°$, $\alpha_{95} = 7.6°$) from D1 (Figure 4b). Five unbaked samples (collected at ~20 to ~160 cm from the contact) resulted in a mean direction of $D = 25.0°$, $I = 33.0°$ ($\alpha_{95} = 3.2°$) close to the group mean direction ($D = 27.6°$, $I = 24.7°$, $\alpha_{95} = 2.3°$, $N = 67$ dikes) reported for the normally magnetized Matachewan dikes (Evans & Halls, 2010; Figure 4b).

Table 3
Summary of Paleomagnetic Results from the Lamprophyre Dikes (This Study) and Abitibi Dikes

Site	λ_S (°N)	Φ_S (°W)	Method	Q	n/N or B(N)	D (°)	I (°)	α_{95} (°)	k	λ_P (°N)	Φ_P (°E)	W (m)	Trend
Reversed polarity dikes													
ML2	48.8112	86.7228	AF + T	A	12/6	103.8	−61.4	5.5	125.0	38.6	206.9	0.3	NNW
D15	48.8027	86.7095	AF + T	A	9/5	126.2	−59.4	10.2	46.2	51.5	191.2	0.8	NW
D16 H	48.8034	86.7066	AF + T	A	11/8	118.8	−57.2	7.6	47.2	45.4	192.9	0.2	NW
MM3	48.8081	86.7076	AF + T	A	6/4	112.3	−85.9	9.9	65.0	51.3	261.2	1.0	E
D14	48.8024	86.7115	AF + T c	B	9/6	124.7	−67.4	9.8	48.0	54.9	207.1	0.4	NW
MM1	48.8029	86.7080	T	B	3/3	106.3	−69.8	12.9	61.6	45.7	218.8	0.2	NW
D10 H	48.8022	86.7145	AF + T c	B	9/7	184.6	−71.4	11.4	45.9	82.2	292.6	1.2	NW
D11	48.8022	86.7145	AF + T c	B	10/6	147.7	−55.6	11.7	25.5	63.3	167.1	0.1	NW
D12	48.8021	86.7141	AF + T c	B	10/6	123.1	−64.9	14.0	19.6	52.7	202.5	0.3	NW
MM2	48.8029	86.7080	AF + T	B	3/2	110.8	−59.7	13.3	58.4	41.8	200.9	0.2	NW
MM2 Baked			AF + T	—	2/1	170.7	−82.4	—	—	—	—	—	—
MM2 Unbaked			AF + T	—	2/1	74.7	86.0	—	—	—	—	—	—
D1	47.7043	83.3191	AF + T c	B	18/8	182.7	−78.1	7.6	46.6	70.5	279.8	0.3	NE
D1 Baked			AF + T	—	4/2	60.9	−75.6	7.9	503.0	—	—	10.0	N
DI Unbaked			AF + T	—	10/5	25.0	33.0	3.2	465.0	—	—	10.0	N
D3 H	47.7041	83.3187	AF + T c	B	5/5	130.0	−68.1	14.2	76.0	57.9	211.9	0.2	NE
D4	48.0584	82.7135	AF + T c	B	18/15	117.9	−80.4	10.7	20.2	53.6	248.8	1.0	NE
D7 H	48.0373	84.5434	AF + T c	B	12/8	102.9	−79.0	14.0	13.7	48.3	243.4	0.7	ENE
Mean for R-polarity dikes					**14(90)**	**126.5**	**−69.9**	**6.3**	**41.3**	**57.5**	**216.6**	**A_{95} = 10.2, K = 16.0**	
A4a[a]	48.45[b]	81.10[b]	AF		2(14)	125.9	−71.7			57.0	223.7		
A7abi[a]	47.08[b]	83.75[b]	AF		2(12)	150.4	−80.8			61.5	257.8		
Grand mean for R-polarity dikes					**16(116)**	**127.2**	**−70.8**	**5.6**	**44.6**	**58.1**	**219.3**	**A_{95} = 9.2, K = 17.2**	
Normal polarity dikes													
D2 H	47.7042	83.3182	AF + T	A	5/5	259.4	81.1	8.6	64.7	42.0	253.4	0.1	NE
D5	48.0590	82.7131	AF + T	A	9/5	331.6	77.2	14.8	11.7	67.0	247.1	0.6	NE
D9	48.0373	84.5434	AF + T	A	10/6	271.2	61.6	14.3	19.0	31.0	216.5	0.2	NE
D13	48.8023	86.7120	AF + T	A	14/11	301.1	81.7	14.6	9.8	54.8	248.7	0.2	NW
ML1	48.8112	86.7228	AF + T	A	8/4	9.5	70.5	11.2	51.1	81.7	314.6	0.3	NNW
Mean for N-polarity dikes					**5(31)**	**305.9**	**78.3**	**13.7**	**32.4**	**57.3**	**242.4**	**A_{95} = 23.8, K = 11.3**	
A3-abi[a]	48.20[b]	84.80[b]	T		2(12)	296.1	54.9			42.5	194.4		
A5-abi[a]	48.40[b]	81.20[b]	AF		4(25)	310.1	59.5	10.7	74.0	54.5	196.0		
A6-1-a[a]	47.90[b]	82.40[b]	AF		5(39)	283.3	56.6	6.5	138.8	35.2	209.2		
A6-2-a[a]	48.50[b]	79.50[b]	AF		7(76)	302.0	71.0	4.4	193.3	54.9	223.3		
A8-abi[a]	48.50[b]	78.20[b]	AF		5(19)	266.3	58.5	9.5	66.4	26.8	222.8		
SL13[c]	48.35[b]	84.67[b]	AF		1(14)	315.9	62.7	5.1	61.0	59.6	194.1		
Abitibi N mean					**6(185)**	**294.7**	**61.6**	**8.9**	**58.0**	**46.3**	**207.7**	**A_{95} = 13.5, K = 25.7**	
Grand mean for N-polarity dikes					**11(216)**	**297.6**	**69.2**	**8.4**	**30.5**	**52.4**	**221.0**	**A_{95} = 13.2, K = 13.0**	
Combined mean (R + N), Lamprophyres					**19(121)**	**306.4**	**72.1**	**5.5**	**37.8**	**58.0**	**223.3**	**A_{95} = 9.2, K = 14.2**	
Combined mean (R + N)					**27(332)**	**303.1**	**70.2**	**4.5**	**38.4**	**55.8**	**220.0**	**A_{95} = 7.3, K = 15.5**	

Note. λ_S, Φ_S: The site latitude and longitude. Method: demagnetization method used (AF = alternating field; T = thermal, or AF + T = combined), "c" denotes entries that include circle-fitted data. Q: The quality class (see text); n/N or B (N): the number of specimen/samples or sites (samples) included in the calculation of the site/group mean declination (D) and inclination (I); α_{95} is the 95% confidence circle and *k* is the concentration parameter (Fisher, 1953). The virtual geomagnetic pole (VGP) has latitude λ_P and longitude Φ_P, A_{95} is the 95% confidence circle and *K* is the concentration parameter. W and Trend: The width and trend of the dike. Bold represents mean directions.
[a]Abitibi dike data taken from Ernst and Buchan (1993). [b]Approximate site coordinates calculated from multiple site locations of the dikes described in the original paper. [c]Abitibi dike data taken from Halls et al. (2005).

The other test was performed on R-polarity dike MM2 that cut the Schreiber-White River Archean greenstone belt rocks in the Marathon area. The ChRM direction (D = 170.7°, I = −82.4°) from a baked country rock sample (taken at 5 cm from the contact) is very broadly similar to the mean direction calculated from two lamprophyre samples (D = 110.8°, I = −59.7°, α_{95} = 13.3°) but is significantly different from the ChRM direction (D = 74.7°, I = 86.0°) obtained from the unbaked sample (Table 3). However, because of the small number of samples, we consider this test as only tentatively positive.

The polarities are symmetrical, that is, the group means for the R- and N-polarity lamprophyre dikes are statistically indistinguishable (Table 3 and Figure 4e), passing the reversal test of McFadden and McElhinny (1990) with C classification (γ_o = 8.4° and γ_c = 14.3°).

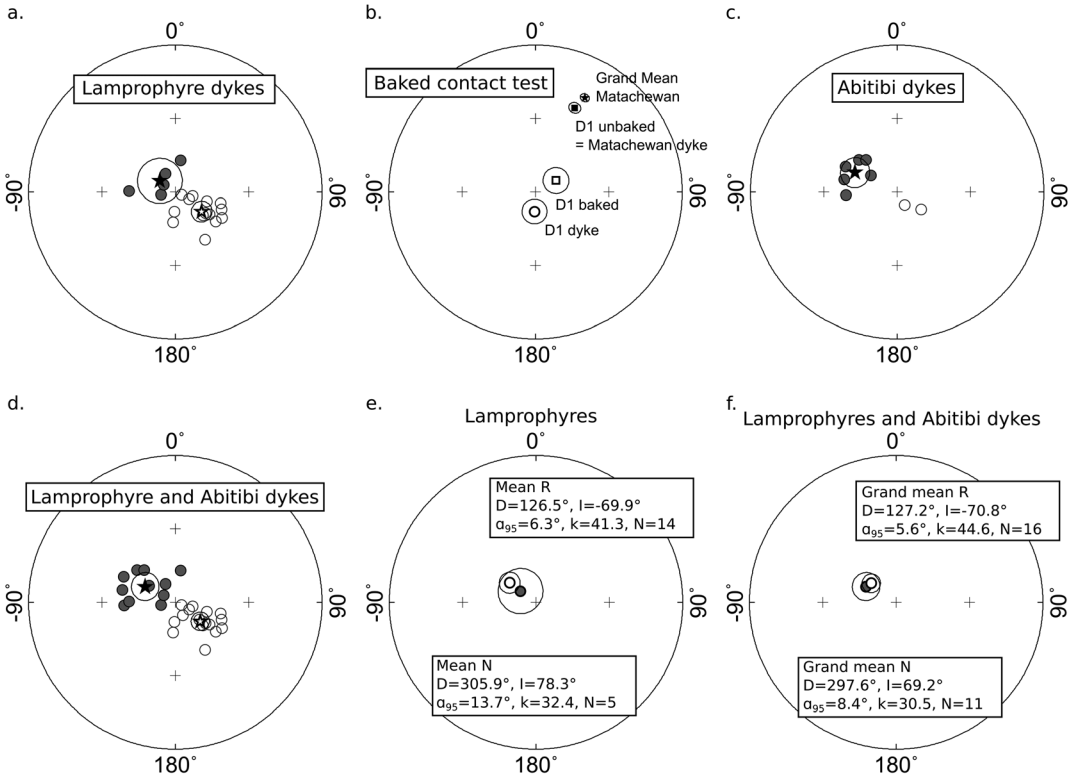


Figure 4. Paleomagnetic results from the lamprophyre and Abitibi dikes. Open (filled) symbols in the equal area plots refer to reversed (normal) polarity. (a) Accepted site mean directions (circles) from the lamprophyre dikes together with the group mean directions (stars) and the 95% confidence circles. (b) The baked contact test for dike D1 (see text). Open circle and square show the mean directions from the lamprophyre dike and baked Matachewan dike, respectively. Filled square shows the mean direction from unbaked Matachewan dike samples. The star illustrates the grand mean for the N-polarity Matachewan dikes (Evans & Halls, 2010). (c) Site mean directions (circles) for the Abitibi dikes (Ernst & Buchan, 1993; Halls et al., 2005) and the group mean direction (star) for the normally magnetized dikes. (d) The combined (from a and c) site mean directions (circles). Stars show the group mean directions calculated for the combined data sets. (e, f) The group mean N-polarity and R-polarity directions and their 95% confidence circles for (e) the lamprophyre dikes (this study) and (f) the combined lamprophyre and Abitibi (Ernst & Buchan, 1993; Halls et al., 2005) data sets. The R-polarity direction has been inverted to their antipodal positions.

4. Discussion

4.1. Geochemistry

The results of our geochemical analyses support the cogenetic nature of the lamprophyre dikes from the Marathon and ELS areas, consistent with their indistinguishable ages (Table 1; Queen et al., 1996). No correlation was observed between the geochemical properties and geomagnetic polarity. Overall, the lamprophyre dikes exhibit a strong affinity with magmas derived from ocean island basalt-type sources (Figure 2e). The high LOI values in the lamprophyres reflect not only the degree of secondary alteration but also the high volatile content (CO_2 and H_2O) of the silicate melt, typical for carbonate-rich ultramafic lamprophyres (Platt et al., 1983; Rock, 1987; Tappe et al., 2008). The CO_2-rich fluid that is responsible for the observed alteration of pyrites and olivines was most probably exsolved during the cooling of the carbonate-rich silicate melt, thus making this alteration effectively synchronous with crystallization of the lamprophyre dikes, that is, primary deuteric alteration and therefore distinct from the secondary alteration.

The lower LOIs and mobile element contents of the Abitibi dikes compared to the lamprophyre dikes may reflect a higher degree of partial melting of the subcontinental lithospheric mantle as its thermal maturation progresses. This would be consistent with emplacement of the Abitibi dikes after the lamprophyre dikes. However, with the indistinguishable radiometric ages (Table 1) and in the absence of any known crosscutting relationships between the lamprophyre and Abitibi dikes, their relative age remains unknown.

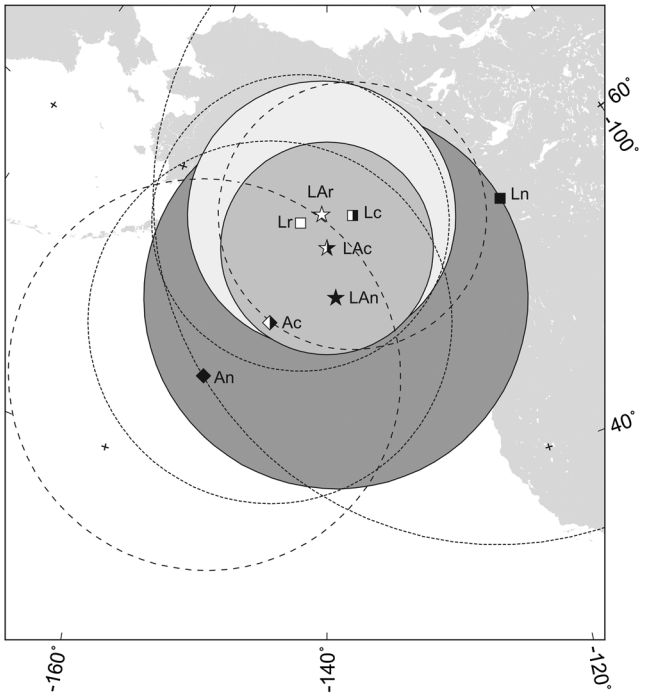


Figure 5. Paleomagnetic poles with the 95% confidence circles (A_{95}) calculated from the lamprophyre and Abitibi dikes (Tables 3 and 4). Solid, open, and half-filled symbols indicate normal, reversed, or mixed polarity. Stars show the new poles from the combined data from the lamprophyre and Abitibi dikes. Square show the poles calculated from the lamprophyre data only. Diamonds show the Abitibi dike poles calculated from the data of Ernst and Buchan (1993) and Halls et al. (2005).

4.2. The Reliability of the Paleomagnetic Directions

In general, the class A paleomagnetic directions were obtained from less altered/metasomatized samples with higher MgO and heavy REE, and lower CaO and LOI contents. In contrast, the dikes that yielded poor quality or not interpretable paleomagnetic data were strongly metasomatized and weathered. The primary remanence in these dikes could have been corrupted by the conversion of magnetite to limonite promoted by the CO_2 metasomatism (DeWitt et al., 1996). We observed no correlation between the quality of paleomagnetic data and the rock magnetic characteristics (magnetic hysteresis parameters and thermomagnetic behavior) of the samples.

The lamprophyre dikes are unlikely to be affected by magnetic overprinting associated with the younger MCR-related intrusions and attendant hydrothermal activity (e.g., Costanzo-Alvarez et al., 1993; Ernst & Buchan, 1993). In the Marathon area, our sampling sites are located well outside of the thermal aureole of the ~1108-Ma Coldwell Complex (Heaman & Machado, 1992; Walker, 1967; Figure 1), and the two N-polarity directions (from ML1 and D13 sites) have significantly steeper inclinations (Table 3) than the remanence (D = 298.0°, I = 56.9°, α_{95} = 5.8°) carried by the western intrusive center of the complex (Kulakov et al., 2014). The dominance of steep R-polarity directions in the Marathon area dikes and the tentatively positive baked contact test for site MM2 further indicate that the intrusion of the Coldwell Complex did not affect their primary remanence.

In the ELS area, the hydrothermal activity associated with emplacement of the 1100.6 ± 1.5-Ma Lackner Lake, 1105.4 ± 2.6-Ma Nemegosenda (Heaman & Machado, 1992) and 1142.6 ± 1.6-Ma Firesand River (Rukhlov & Bell, 2010) carbonatite complexes could have created a steep R-polarity thermochemical overprint in some adjacent rocks (Costanzo-Alvarez et al., 1993). However, our sampling sites are located at distances (>10 km) greatly exceeding the extent of alteration aureole of the complexes. More importantly, the primary origin of ChRM in the ELS area lamprophyres is reliably established by the positive baked contact test for dike D1 (Figure 4b).

Finally, the primary origin of ChRM in the lamprophyre dikes from both ELS and Marathon areas is further reinforced by the positive reversal test for the group mean directions calculated from the normally and reversely magnetized dikes.

4.3. Comparison With the Abitibi Dikes

Both reversed polarity directions from the Abitibi dikes plot within the scatter of R-polarity directions from the lamprophyres, yielding the combined group mean direction of D = 127.2°, I = −70.8° (α_{95} = 5.6°, N = 16; Figure 4d and Table 3). The group means for normal polarity directions from the lamprophyre dikes (D = 305.9°, I = 78.3°, α_{95} = 13.7°, N = 5) and Abitibi dikes (D = 294.7°, I = 61.6°, α_{95} = 8.9°, N = 6) are statistically indistinguishable (Figures 4a and 4c). The combined N-polarity group mean (D = 297.6°, I = 69.2°, α_{95} = 8.4°, N = 11) passes the reversal test with the combined R-polarity group mean with B classification (γ_o = 3.6° and the γ_c = 9.9°; Figure 4f). The respective paleomagnetic poles (LAn: 52.4°N, 221.0°E, A_{95} = 13.2°) and (LAr: 58.1°N, 219.3°E, A_{95} = 9.2°) are also indistinguishable (Figure 5). The positive baked contact test for the N-polarity Abitibi dikes (Ernst & Buchan, 1993) provides further support for reliability of the combined poles.

The angular dispersion (S) of virtual geomagnetic poles (VGPs) was calculated to assess if the paleomagnetic data sets adequately sample the paleosecular variation:

$$S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\Delta_i^2 \quad (1)$$

Here N is the number of individual VGPs and Δ_i denotes the ith VGP to group mean pole angular distance. The scatter of the values was corrected for within-site dispersion (S_w) corresponding to intrinsic variation and experimental uncertainty by using a formula:

$$S_b^2 = S^2 - S_w^2/n, \quad (2)$$

where S_b is the true (between-site) dispersion and n is the average number of samples per site (Doell, 1970). We used the N-1 jackknife method (Efron, 1982) to estimate the 95% confidence interval of S_b.

The values calculated from the combined R-polarity data set (S_b = 18.4 ± 6.0°, N = 16) and from the lamprophyre dikes only (S_b = 18.7 ± 5.8°, N = 14) are identical and in a good agreement with the long-term value (S_b = 16.0 ± 4.5°) expected at the 50–55° latitude for the 1.0–2.2-Ga period (Smirnov et al., 2011). The value calculated from the combined N-polarity data set (S_b = 22.2 ± 3.7°, N = 11) is somewhat higher than the expected value but is still consistent within the uncertainty. The value S_b = 20.0 ± 4.4° (N = 27) calculated from the combined data set of both polarities is also within the expected range of secular variation. The application of a variable cutoff (Vandamme, 1994) does not affect the calculated S_b values, indicating that none of the accepted VGPs represents transitional field. Overall, our analysis of the VGP angular dispersion indicates that the combined data sets for both R-polarity and N-polarity directions represent the time-averaged field.

The observed similarities in age, local trends, geochemistry, and paleomagnetism between the lamprophyre and Abitibi dikes indicate that both suites likely represent a single magmatic pulse or magmatic events closely spaced in time. The combined mean remanence direction (D = 303.1°, I = 70.2°, α_{95} = 4.5°, N = 27) corresponds to a grand mean paleomagnetic pole (Pole LAc: 55.8°N, 220.0°E, A_{95} = 7.3°, N = 27; Table 3 and Figures 5 and 6). The new pole merits the highest seven-point classification on Van der Voo's (1990) paleomagnetic reliability scale (Q-scale) and represents a key paleomagnetic pole for Laurentia at ~1143 ± 2 Ma.

4.4. Implications for the North American Plate Motion

Well-defined APWPs based on high-quality paleomagnetic poles are crucial for paleocontinental reconstructions as well as for constraining the rates of plate motion and true polar wander (TPW; e.g., Besse & Courtillot, 2002, 2003; Steinberger & Torsvik, 2008). In our subsequent discussion, we used the PALEOMAGIA database (Veikkolainen et al., 2017) to select the most reliable paleomagnetic poles with their quality factor (1–6) greater than or equal to 5 (Table 4 and Figure 6).

The North American APWP for the late Mesoproterozoic (~1300–1050 Ma) is characterized by two nearly parallel, long swathes of polar motion from equatorial to high latitudes between ~1300 and ~1160 Ma and back to equatorial latitudes by ~1050 Ma (Figure 6 and Table 4). The swathes form an elongate loop often referred to as the Logan Loop (Robertson & Fahrig, 1971). The southern end of the ascending swathe, the Mackenzie-Gardar track (e.g., Piper, 2010), is anchored by the equatorial poles from the ~1267-Ma Mackenzie igneous province and the ~1235-Ma Sudbury dike swarm. Several igneous and sedimentary rock suites comprising the Mesoproterozoic Gardar Igneous Province of South Greenland (e.g., Upton, 2013) record the subsequent northward migration of Laurentia. The northeastern end of the Gardar polar wander track is defined by the ~1160-Ma Ilímaussaq Complex (Piper et al., 1999).

The younger, descending swathe of the loop, the Keweenawan track, is defined by the poles from rocks of the MCR (e.g., Ojakangas et al., 2001) and the Southwestern Laurentia large igneous province (Bright et al., 2014; Figure 6 and Table 4). However, the northern extremity of this track is not well constrained. The oldest pole from the Abitibi dikes (Ernst & Buchan, 1993; Halls et al., 2005) plots close to the poles representing the first main pulse of MCR magmatism at ~1105–1115 Ma (Figure 6). However, these poles are rather scattered and only two poles have been precisely dated—one from the reversed rocks of Coldwell Complex (1108 ± 1 Ma; Heaman & Machado, 1992) and the other from the upper part of Osler volcanics (1105 ± 2 Ma; Davis & Green, 1997; Table 4). Another pole from reversely magnetized diabase sheets of Central Arizona dated at 1119 ± 10 Ma (Donadini et al., 2011, 2012) plots noticeably to the east from the cluster of other 1105- to 1115-Ma poles (Figure 6). The ages of other poles have been assigned based on geological or/and

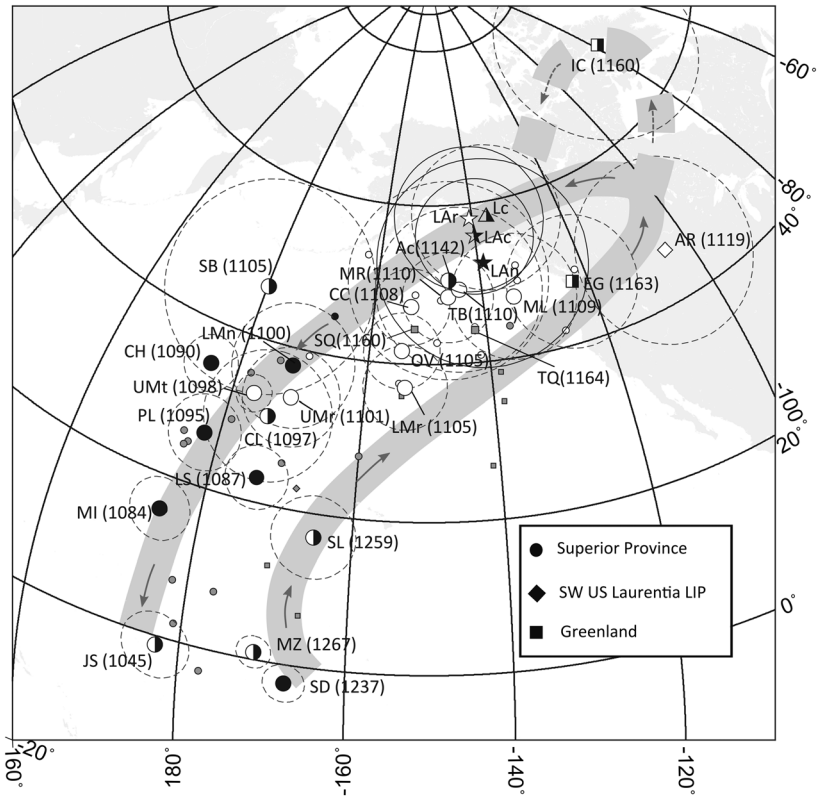


Figure 6. Selected Laurentian paleomagnetic poles obtained from rocks of the Superior province (circles), Southwestern U.S. LIP (diamonds), and Greenland (squares) for ~1300–1000 Ma (see text and Table 4). Solid, open, and half-filled symbols indicate normal, reversed, or mixed polarity. Small symbols without the A_{95} circles and larger symbols with A_{95} circles show the poles with $QF = 4$ and $QF \geq 5$, respectively. The abbreviations are the same as in Table 4. Stars show the new combined paleomagnetic poles from the lamprophyres and Abitibi dikes. The triangle shows the combined pole calculated from the lamprophyre data only. The thick gray line indicates the apparent polar wander path (the Logan Loop) with a potential northward extension shown by the thick dashed line. The dark gray arrows indicate the direction of time.

paleomagnetic correlations (e.g., Veikkolainen et al., 2017). The Abitibi pole is statistically indistinguishable from the R-polarity Coldwell Complex pole (47.2°N, 206.5°E, A_{95} = 4.8°; Kulakov et al., 2014) suggesting a very slow or no latitudinal motion of Laurentia between ~1142 and ~1108 Ma (0.17 ± 0.20°/Myr). However, the Abitibi pole does not represent the time-averaged field, and therefore, the reconstructions based on this pole are likely to be biased. Our combined grand pole from the Abitibi and lamprophyre dikes plots about 6° northeast from the Abitibi pole (Figures 5 and 6). The new pole reliably constrains the northeastern extremity of the Keweenawan track, suggesting a more northerly position of the craton than was previously indicated by the Abitibi pole.

We calculated the apparent polar wander (APW) rate using our new pole and the R-polarity Coldwell Complex pole. The estimate of APW rate at 3.8 ± 1.4 cm/year (0.34 ± 0.13°/Myr) is similar to the rate of latitudinal motion of 3.4 ± 1.4 cm/year obtained from the same data sets, which indicates that, between ~1143 and ~1108 Ma, the craton experienced mainly southward latitudinal motion with no or negligible longitudinal component. The calculated plate velocity is comparable with the present-day velocities for large continental plates (e.g., DeMets et al., 1990). We note that the significant scatter of the ~1108- to 1143-Ma poles may allow for higher velocities during this period. However, even if our ~1142-Ma pole is averaged with the ~1119-Ma pole from Central Arizona diabases, the resulting motion rate of ~6–7 cm/year is still within the "normal" range.

Interestingly, the ~35-Ma-long period of normal plate rate (~1143–1108 Ma) was immediately followed by the period of an unusually fast equatorward plate movement with the rate of at least 24.6 ± 3.8 cm/year (2.21 ± 0.34°/Myr) between ~1108 and ~1099 Ma (Kulakov et al., 2014; Swanson-Hysell et al., 2009).

Table 4
Selected Paleomagnetic Poles for Laurentia and Greenland at ~1240–1045 Ma

Rock unit	Abbr.	B	λ_p	φ_p	A_{95}(°)	QF	Age (Ma)	Max-min age (Ma)	References
Laurentia									
Lamprophyre mean (R)	Lr	14	57.5	216.6	10.2	5	1144	1150–1140[a]	This study
Lamprophyre mean (N)	Ln	5	57.3	242.4	23.8	5	1144	1150–1140[a]	This study
Lamprophyre combined (M)	Lc	19	58.0	223.3	9.2	6	1144	1150–1140[a]	This study
Lamprophyre + Abitibi grand mean (N)	LAn	11	52.4	221.0	13.2	5	1143	1150–1139[a]	This study
Lamprophyre + Abitibi grand mean (R)	LAr	16	58.1	219.3	9.2	5	1143	1150–1139[a]	This study
Lamprophyre + Abitibi combined (M)	LAc	27	55.8	220.0	7.3	6	1143	1150–1139[a]	This study
Mackenzie diabase (M)	MZ	35	1	189	2.1	6	1267	1269–1265[a]	Irving et al. (1972)
Seal Lake group igneous rocks (M)	SL	11	17	195	5.3	5	1259	1274–1243[a]	Park and Roy (1979)
Sudbury dikes (N)	SD	52	−2.5	192.8	2.5	5	1237	1242–1232[a]	Palmer et al. (1977)
Abitibi dikes (N)	An	6	46.1	207.2	13.5	5	1142	1143–1139[a]	Ernst and Buchan (1993) and Halls et al. (2005)
Abitibi dikes (M)	Ac	8	44.4	211.4	13.5	6	1142	1143–1139[a]	Ernst and Buchan (1993) and Halls et al. (2005)
Central Arizona diabases grand mean (R)	AR	5	46.2	255.5	11.5	5	1119	1129–1103[a]	Donadini et al. (2011)
Thunder Bay sills and baked rocks (R)	TB	10	49.3	215.7	10.4	5	1110	1130–1090	Pesonen (1979)
Marquette dikes (R)	MR	14	48.4	213.5	5.7	5	1110	1120–1100	Pesonen (1979)
Lowermost Mamainse lavas (R)	ML	14	47.6	225.8	7.9	5	1109	1120–1100	Swanson-Hysell et al. (2009)
Coldwell Complex Centers A and C (R)	CC	30	47.2	206.7	4.8	5	1108	1112–1102[a]	Kulakov et al. (2014)
Seabrook Lake carbonatite complex (M)	SB	13	46.3	180.2	13.4	5	1105	1149–1077[a]	Symons (1992)
Lower Mamainse lavas (R)	LMr	12	37.0	206.2	5.3	5	1105	1110–1100	Swanson-Hysell et al. (2009)
Upper third Osler volcanics (R)	OV	34	41.6	205.4	4.8	5	1105	1107–1103[a]	Swanson-Hysell et al. (2014)
Lower Mamainse Point volcanics (N)	LMn	14	37.8	188.1	7.8	5	1100	1101–1099	Swanson-Hysell et al. (2014)
Upper Mamainse lavas (R)	UMr	10	33.9	188.9	6.2	5	1100	1100.4 ± 0.3[a]	Swanson-Hysell et al. (2009)
Uppermost Mamainse lavas (N)	UMt	21	33.2	183.3	2.3	5	1098	1100–1095	Swanson-Hysell et al. (2009)
Chengwatana lavas (M)	CL	8	30.9	186.1	8.2	5	1095	1097–1093[a]	Kean et al. (1997)
Portage Lake volcanics (N)	PL	28	26.6	177.9	4.7	5	1095	1098–1093[a]	Hnat et al. (2006)
Copper Harbor lavas (N)	CH	7	34.9	175.6	3.5	5	1090	1095–1085	Halls and Palmer (1981)
Lake Shore Traps (N)	LS	23	23.1	186.4	4.0	5	1086	1089–1085[a]	Kulakov et al. (2013)
Michipicoten Island formation (N)	MI	21	15.7	174.7	3.9	5	1084	1085–1083[a]	Fairchild et al. (2017)
Jacobsville formation (M)	JS	27	−1.2	177.1	4.2	5	1045	1050–990	Roy and Robertson (1978)
Greenland									
Ilimaussaq Complex (M)	IC	10	69.2	281.2	12.2	6	1160	1165–1155[a]	Piper et al. (1999)
Eriksfjord group (M)	EG	3	47.6	237.1	8.3	5	1164	1167–1161[a]	Piper et al. (1999)
Giant gabbro dikes of Tugtutôq (N)	TQ	13	44.1	218.0	7.6	4	1164	1167–1161[a]	Piper (1977)
South Qôroq intrusion (N)	SQ	9	44.4	207.4	12.0	4	1160	1168–1152[a]	Piper (1992)

Note. The poles that have the quality factor $QF \geq 5$ were selected from the PALEOMAGIA database (Veikkolainen et al., 2017). N/R/M denote normal/reversed/mixed polarity. Abbr = abbreviation; B = the number of sites; λ_p and φ_p = VGP pole latitude and longitude; A_{95}(°) = radius of 95% confidence circle; QF = the quality factor (1–6) as described in Veikkolainen et al. (2017); Age = the nominal age assigned to the pole in the PALEOMAGIA database; Max-min age = the upper and lower limit of the age estimate. The Greenland poles are rotated −13.8° to Laurentian coordinates around an Euler pole at 67.5°N, 241.5°E (Roest & Srivastava, 1989).
[a]Denotes that the age estimate is based on radiometric dating.

Kulakov et al. (2014) hypothesized that the ultrafast motion of Laurentia could have been facilitated by a reduced asthenospheric resistance due to unusually vigorous mantle suggested by extensive intraplate magmatism at ~1195–1115 Ma. However, the authors did not rule out TPW (e.g., Evans, 2003) as a potential explanation. The sudden acceleration of the plate at ~1108 Ma is consistent with both mechanisms.

An additional insight into this problem may be gained from the pre-1143-Ma data. Presently, the apex of the Logan Loop is defined by a single pole from the marginal syenites of the alkaline Ilímaussaq Complex in South Greenland (Piper et al., 1999), which plots about 30° NE from our new pole (Table 4 and Figure 6). However, the age of the complex has not been reliably established. The published age determinations include 1143 ± 21 Ma (Rb-Sr; Blaxland et al., 1976) and 1130 ± 50 Ma (Sm-Nd; Paslick et al., 1993). Krumrei et al. (2006) reported an U-Pb baddeleyite age of 1160 ± 5 Ma representing the earliest magmatic pulse (augite syenite), which is consistent with the zircon age of 1166 ± 9 Ma from the second stage granite cited in Upton et al. (2003) as a work in preparation. Krumrei et al. (2006) also reported ^{40}Ar-^{39}Ar ages ranging between 1152.3 ± 3.7 and 1142.6 ± 2.2 Ma obtained from amphiboles from both earliest and latest magmatic stages. Finally, based on in situ laser ablation isotopic analyses of eudialyte, Wu et al. (2010) reported U-Pb

ages ranging from 1134 ± 29 to 1018 ± 34 Ma, a Sm-Nd isochron age of 1028 ± 63 Ma, and a Hf-Lu isochron consistent with an ~1160-Ma age. The upper limit for the age is constrained by the estimate of 1134 ± 17 Ma (Rb-Sr) from a fresh monchiquite dike that cuts the Ilímaussaq intrusion (Larsen, 2006).

Although the geochronological data allow for the Ilímaussaq Complex to be similar in age to the lamprophyre and Abitibi dikes, this would imply an unrealistically fast plate velocity for Laurentia. The estimate at 1160 ± 5 Ma (Krumrei et al., 2006) has been recently cited as the most plausible age of the complex (e.g., Upton, 2013). Based on this age, the calculated minimal plate velocity for the 1160- to 1143-Ma period (19.4 ± 5.6 cm/year; 1.74 ± 0.50 °/Myr) is comparable with the ultrafast velocity reported for the 1108- to 1099-Ma period (Kulakov et al., 2014).

Therefore, taken at face value, the available geochronological and paleomagnetic data suggest two dramatic changes closely spaced in time in the rate of North American plate: from an unusually rapid motion (~1160–1143 Ma) to a 5–6 times slower rate for 1143–1108 Ma and back to an ultrafast rate (~1108–1099 Ma). If robust, this observation represents a geodynamical conundrum. However, such a scenario ultimately depends on the validity of the Ilímaussaq Complex pole. We note that the coeval ~1160-Ma pole from the Eriksfjord Group red beds (Piper et al., 1999) plots significantly south of the Ilímaussaq Complex pole (Figure 6). However, this pole is based on the overprint remanence from the dikes and plutons emplaced during the circa 1160-Ma late stage of the Gardar magmatic activity and has only been dated by paleomagnetic correlation (Piper et al., 1999). The PALEOMAGIA database contains two reliably dated late Gardar poles: the pole from the South Qôroq intrusion (Piper, 1992) dated at 1160 ± 8 Ma (Blaxland et al., 1978) and the pole from the Younger Giant Dyke Complex of Tugtutôq (Piper, 1977) dated at 1163 ± 2 Ma (U-Pb; Buchan et al., 2001; Table 4). Although these poles have a slightly lower quality rating (quality factor = 4), they plot within the cluster of other Laurentian poles for ~1108–1163 Ma (Figure 6). Assuming that these poles have only a few million year age differences with the Ilímaussaq Complex pole, the results infer unrealistically high plate velocity estimates. In fact, any age of the Ilímaussaq Complex between 1160 and 1143 Ma results in an implausible geodynamic scenario with the minimal plate velocity exceeding 45 cm/year and involving a sudden inversion in the direction of plate motion.

The Ilímaussaq Complex pole therefore seems to be anomalous and warrants additional scrutiny. Although such an endeavor is outside of the scope of this paper, we remind that the paleomagnetic directions determined from the syenites are indistinguishable from the present-day field direction. Piper et al. (1999) interpreted these directions as primary because of the similarity of directions from two different locations, high unblocking temperatures, and the presence of a steep inclination overprint in some adjacent lava flows of Eriksfjord Formation (although the overprint declinations were not consistent). However, these observations do not preclude a possibility of recent overprinting by lightning, which may result in very stable remanence (Dunlop & Özdemir, 1997). Alternatively, the existence of a long-lived, substantially nondipole geomagnetic field at ~1.1 Ga has been hypothesized (Donadini et al., 2011; Pesonen & Nevanlinna, 1981), which could be responsible for the unusual position of the Ilímaussaq Complex and Central Arizona poles. However, such a possibility is less likely in the view of recent results indicating a largely dipolar geometry of the Proterozoic field (Evans, 2006; Kulakov et al., 2014; Smirnov et al., 2011; Swanson-Hysell et al., 2009).

If the Ilímaussaq Complex pole is excluded, the positions of remaining poles suggest that, between ~1108 and ~1163 Ma, the craton was positioned at midlatitudes and moved with a velocity characteristic for large continental plates today. Furthermore, the location of ~1160-Ma poles south of our new pole (Figure 6) suggests a measurable northward component of the craton's motion during that time. As a very crude estimate, we used our new pole and the pole from the Younger Giant Dyke Complex of Tugtutôq (Piper, 1977) to calculate the APW rate of 6.5 ± 3.3 cm/year (0.59 ± 0.30°/Myr) between ~1163 and ~1143 Ma. We also obtained a similar APW velocity (5.4 ± 0.6 cm/year or 0.59 ± 0.30°/Myr) using the Tugtutôq pole and the pole from the Mackenzie diabase province (Irving et al., 1972) dated at 1267 ± 2 Ma (LeCheminant & Heaman, 1989). Both estimates are significantly slower than the maximum APW rates for Laurentia and Gondwana in the Phanerozoic (e.g., Meert et al., 1993).

Based on these considerations, our preferred hypothesis is that the northward motion of Laurentia from its equatorial position at ~1250 Ma occurred with velocities typical for large continental plates and did not proceed to the high latitudes suggested by the Ilímaussaq Complex pole. Instead, after reaching the midlatitude range at ~1160–1170 Ma, the plate slowed down and/or changed the direction of its motion to contain a

longitudinal or rotational component, possibly, as a result of the second episode of Gardar rifting. The plate continued to move with a normal velocity, changing its direction to mainly equatorward at ~1143 Ma, possibly due to the commencement of Midcontinent rifting recorded by the lamprophyre dikes and Abitibi dikes. This motion continued until a sudden acceleration at ~1108 Ma contemporaneous with the first main pulse of MCR magmatism (1115–1105 Ma), which could reflect a reduced mantle resistance due to abnormally hot and active mantle (Kulakov et al., 2014), or a TPW event (Evans, 2003), or a combination of these.

4.5. Implications for the Genesis of the MCR

The Marathon-ELS lamprophyre and Abitibi dikes as well as the younger alkali-carbonatitic complexes are concentrated within preexisting crustal weakness zones (such as the Kapuskasing Structural Zone and the Trans-Superior Tectonic Zone) near the MCR rift center (Figure 1). Tappe et al. (2008) suggested that the production of carbonatite rich magmas is induced solely by thinning of the overlying cratonic lithosphere, which reactivates the preexisting phlogopite-carbonatite dominated metasomatized veins in the lithosphere. The thinning does not necessary require a mantle plume, but the impinging mantle plume head can also produce extensional stresses on the overlying lithosphere. The Keweenawan alkaline and carbonatite magmatism in both of these areas have long been suggested to be a manifestation of a failed arm related with the impinging mantle plume (Ernst & Bell, 2010; Platt & Mitchell, 1979). In particular, Ernst and Bell (2010) suggested that all ~1.1-Ga carbonatite complexes in these areas represent the extent of the plume head with a radius of ~1,000 km. Based on the similar ages between the carbonatites and the ~1143-Ma lamprophyres as well as the similar carbonate-rich geochemical nature and, especially, the OIB affinity of the lamprophyres gained from the more reliable trace element data, we suggest that the carbonate-rich lamprophyres may be genetically related to an impinging mantle plume. Irrespective of the origin of rifting, the Marathon-ELS lamprophyre dikes together with the Abitibi dikes represent the incipient magmatism of the MCR as previously suggested by Queen et al. (1996).

5. Conclusions

The ~1144-Ma ultramafic lamprophyre dikes cropping out northeast of Lake Superior (Ontario, Canada) faithfully record the direction of Earth's magnetic field at the time of their formation. The similarity of paleomagnetic directions and geochemical characteristics between the lamprophyre dikes and the coeval Abitibi dikes supports the earlier hypothesis that both dike swarms are connected, representing the earliest magmatic event associated with the commencement of the ~1.1-Ga North American Midcontinent rifting (Queen et al., 1996). The high-quality paleomagnetic pole calculated from the combined paleodirectional data set for the lamprophyre and Abitibi dikes represents one of the key poles defining the northern extremity of the Keweenawan track of the North American APWP. The currently accepted position of the Logan Loop apex, defined by a single paleomagnetic pole from the Ilímaussaq Complex in South Greenland, implies an implausible geodynamic scenario for the plate motion and needs to be taken with caution. The positions of other reliable poles for ~1100–1160 Ma that cluster at a substantial distance southwest from the Ilímaussaq Complex pole suggest that the velocity of Laurentia motion between ~1250 and 1110 Ma was well within the range of the typical velocities reported for large continental plates during the Phanerozoic.

Acknowledgments
We thank the Institute for Lake Superior Geology (ILSG) for their support for the summer 2013 fieldwork (to E. J. P.). Angela Manderfield and Celine Mandon are recognized for helping with the fieldwork and Darcy Jacobson and Jeanine Chiemilevski for helping with the paleomagnetic and rock magnetic measurements. We also thank Chad Deering and John Gamble for discussions and guidance with geochemistry. We thank Sergei Pisarevsky, Anthony Pivarunas, and Douwe van Hinsbergen for their helpful reviews and comments. This research was supported in part by Grant EAR-1149434 from the U.S. National Science Foundation (to A. V. S.). R. H. M.'s work on alkaline rock geochemistry was supported by Natural Sciences and Engineering Research Council of Canada. All data needed to evaluate the conclusions of this study are provided in the paper and the supporting information.

References

Besse, J., & Courtillot, V. (2002). Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr. *Journal of Geophysical Research*, *107*(B11), 246l4. https://doi.org/10.1029/2003JB002684

Besse, J., & Courtillot, V. (2003). Correction to "Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr.". *Journal of Geophysical Research*, *108*(B10), 2469. https://doi.org/10.1029/2003JB002684

Blaxland, A. B., Van Breemen, O., Emeleus, C. H., Anderson, J. G., & Steenfelt, A. (1978). Age and origin of the major syenite centers in the Gardar province of South Greenland: Rb-Sr studies. *Geological Society of America Bulletin*, *89*(2), 231. https://doi.org/10.1130/0016-7606(1978)89<231:AAOOTM>2.0.CO;2

Blaxland, A. B., Van Breemen, O., & Steenfelt, A. (1976). Age and origin of agpaitic magmatism at Ilimaussaq, South Greenland: Rb-Sr study. *Lithos*, *9*(1), 31–38. https://doi.org/10.1016/0024-4937(76)90054-2

Bright, R. M., Amato, J. M., Denyszyn, S. W., & Ernst, R. E. (2014). U-Pb geochronology of 1.1 Ga diabase in the southwestern United States: Testing models for the origin of a post-Grenville large igneous province. *Lithosphere*, *6*(3), 135–156. https://doi.org/10.1130/L335.1

Buchan, K. L., & Ernst, R. E. (2004). Diabase dyke swarms and related units in Canada and adjacent regions, Map 2022A, Accompanying notes. *Geological Survey of Canada*.

Buchan, K. L., Ernst, R. E., Hamilton, M. A., Mertanen, S., Pesonen, L. J., & Elming, S. Å. (2001). Rodinia: The evidence from integrated palaeomagnetism and U-Pb geochronology. *Precambrian Research*, *110*(1–4), 9–32. https://doi.org/10.1016/S0301-9268(01)00178-4

Costanzo-Alvarez, V., Dunlop, D. J., & Pesonen, L. J. (1993). Paleomagnetism of alkaline complexes and remagnetization in the Kapuskasing Structural Zone, Ontario, Canada. *Journal of Geophysical Research*, *98*(B3), 4063–4079. https://doi.org/10.1029/92JB02571

Davis, D. W., & Green, J. C. (1997). Geochronology of the North American Midcontinen rift in western Lake Superior and implications for its geodynamic evolution. *Canadian Journal of Earth Sciences*, *34*(4), 476–488. https://doi.org/10.1139/e17-039

Day, R., Fuller, M., & Schmidt, V. A. (1977). Hysteresis properties of titanomagnetites: Grain-size and compositional dependence. *Physics of the Earth and Planetary Interiors*, *13*(4), 260–267. https://doi.org/10.1016/0031-9201(77)90108-X

DeMets, C., Gordon, R. G., Argus, D. F., & Stein, S. (1990). Current plate motions. *Geophysical Journal International*, *101*(2), 425–478. https://doi.org/10.1111/j.1365-246X.1990.tb06579.x

DeWitt, B. E., Foord, E. E., Zartman, R. E., Pearson, R. C., & Foster, F. (1996). Chronology of late cretaceous igneous and hydrothermal events at the golden sunlight gold-silver breccia pipe, southwestern Montana. *USGS Bulletin*, *2155*, 48.

Doell, R. R. (1970). Paleomagnetic secular variation study of lavas from the Massif Central, France. *Earth and Planetary Science Letters*, *8*(5), 352–362. https://doi.org/10.1016/0012-821X(70)90108-1

Donadini, F., Pesonen, L. J., Korhonen, K., Deutsch, A., & Harlan, S. S. (2011). Paleomagnetism and paleointensity of the 1.1 Ga old diabase sheets from Central Arizona. *Geophysica*, *47*(1–2), 3–30.

Donadini, F., Pesonen, L. J., Korhonen, K., Deutsch, A., Harlan, S. S., & Heaman, L. M. (2012). Symmetric and asymmetric reversals in the 1.1 Ga central Arizona diabases: The debate continues. In S. Mertanen, L. J. Pesonen, & P. Sangchan (Eds.), *Supercontinent Symposium 2012 – Programme and Abstracts* (pp. 35–36). Geological Survey of Finland, Espoo, Finland.

Dunlop, D. J. (2002). Theory and application of the day plot (Mrs/Ms versus Hcr/Hc) 1. Theoretical curves and tests using titanomagnetite data. *Journal of Geophysical Research*, *107*(B3), 2056. https://doi.org/10.1029/2001JB000486

Dunlop, D. J., & Özdemir, Ö. (1997). *Rock magnetism: Fundamentals and frontiers. Cambridge University Pr*. Cambridge, United Kingdom: Cambridge University Press. https://doi.org/10.1017/CBO9781107415324.004

Efron, B. (1982). *The jackknife, the bootstrap and other resampling plans. Regional conference series in applied mathematics* (Vol. 38). Philadelphia: SIAM.

Ernst, R. E., & Bell, K. (2010). Large igneous provinces (LIPs) and carbonatites. *Mineralogy and Petrology*, *98*(1-4), 55–76. https://doi.org/10.1007/s00710-009-0074-1

Ernst, R. E., & Buchan, K. L. (1993). Paleomagnetism of the Abitibi dyke swarm, southern Superior Province, and implications for the Logan Loop. *Canadian Journal of Earth Sciences*, *30*(9), 1886–1897. https://doi.org/10.1139/e93-167

Ernst, R. E., & Buchan, K. L. (2010). Geochemical database of Proterozoic intraplate mafic magmatism in Canada. Geological Survey of Canada, Open File 6016.

Evans, D. A. D. (2003). True polar wander and supercontinents. *Tectonophysics*, *362*(1–4), 303–320. https://doi.org/10.1016/S0040-1951(02)000642-X

Evans, D. A. D. (2006). Proterozoic low orbital obliquity and axial-dipolar geomagnetic field from evaporite palaeolatitudes. *Nature*, *444*(7115), 51–55. https://doi.org/10.1038/nature05203

Evans, D. A. D., & Halls, H. C. (2010). Restoring Proterozoic deformation within the Superior craton. *Precambrian Research*, *183*(3), 474–489. https://doi.org/10.1016/j.precamres.2010.02.007

Fairchild, L. M., Swanson-Hysell, N. L., Ramezani, J., Sprain, C. J., & Bowring, S. A. (2017). The end of Midcontinent Rift magmatism and the paleogeography of Laurentia. *Lithosphere*, *9*(1), 117–133. https://doi.org/10.1130/L580.1. https://doi.org/10.1130/L580.1

Fisher, R. (1953). Dispersion on a sphere. *Proceedings of the Royal Society of London A*, *217*, 295–305.

Goodenough, K. M., Upton, B. G. J., & Ellam, R. M. (2002). Long-term memory of subduction processes in the lithospheric mantle: Evidence from the geochemistry of basic dykes in the Gardar Province of South Greenland. *Journal of the Geological Society*, *159*(6), 705–714. https://doi.org/10.1144/0016-764901-154

Green, J. C., Bornhorts, T. J., Chandler, V. W., Mudrey, M. G., Myers, P. E., Pesonen, L. J., & Wilband, J. T. (1987). Keweenawan dykes of the Lake Superior region: evidence for evolution of the Middle Proterozoic Midcontinent Rift of North America. In H. C. Halls & W. F. Fahrig (Eds.), *Mafic dyke swarms* (pp. 289–302). Geological Association of Canada, Special Paper 34.

Haggerty, S. E. (1991). Oxide textures—A mini atlas. In D. H. Lindsley (Ed.), oxide minerals: Petrologic and magnetic significance. *Reviews in Mineralogy*, *25*, 129–219). Min. Soc. Amer.

Halama, R., Joron, J.-L., Villemant, B., Markl, G., & Treuil, M. (2007). Trace element constraints on mantle sources during mid-Proterozoic magmatism: Evidence for a link between the Gardar (South Greenland) and Abitibi (Canadian Shield) mafic rocks. *Canadian Journal of Earth Sciences*, *44*(4), 459–478. https://doi.org/10.1139/e06-108

Halls, H., Stott, G. M., & Davis, D. W. (2005). Paleomagnetism, geochronology and geochemistry of several Proterozoic mafic dike dike swarms in northwestern Ontario. Ontario Geological Survey Open File Report, 6171, 1–59.

Halls, H. C., & Palmer, H. C. (1981). Remagnetization in Keweenawan rocks. Part II: Lava flows within the Copper Harbor Conglomerate, Michigan. *Canadian Journal of Earth Sciences*, *18*(9), 1395–1408. https://doi.org/10.1139/e81-131

Halls, H. C., & Pesonen, L. J. (1982). Paleomagnetism of Keweenawan rocks. *Geological Society of America Memoirs*, *156*, 173–201. https://doi.org/10.1130/MEM156-p173

Heaman, L. M., Easton, R. M., Hart, T. R., Hollings, P., MacDonald, C. A., & Smyk, M. (2007). Further refinement to the timing of Mesoproterozoic magmatism, Lake Nipigon region, Ontario. *Canadian Journal of Earth Sciences*, *44*(8), 1055–1086. https://doi.org/10.1139/e06-117

Heaman, L. M., & Machado, N. (1992). Timing and origin of midcontinent rift alkaline magmatism, North America: Evidence from the Coldwell Complex. *Contributions to Mineralogy and Petrology*, *110*(2–3), 289–303. https://doi.org/10.1007/BF00310744

Hnat, J. S., van der Pluijm, B. A., & van der Voo, R. (2006). Primary curvature in the Mid-Continent Rift: Paleomagnetism of the Portage Lake Volcanics (northern Michigan, USA). *Tectonophysics*, *425*(1–4), 71–82. https://doi.org/10.1016/j.tecto.2006.07.006

Hollings, P., Smyk, M., Heaman, L. M., & Halls, H. (2010). The geochemistry, geochronology and paleomagnetism of dikes and sills associated with the Mesoproterozoic Midcontinent Rift near Thunder Bay, Ontario, Canada. *Precambrian Research*, *183*(3), 553–571. https://doi.org/10.1016/j.precamres.2010.01.012

Hutchinson, D. R., White, R. S., Cannon, W. F., & Schulz, K. J. (1990). Keweenaw hot spot: Geophysical evidence for a 1.1 Ga mantle plume beneath the Midcontinent Rift System. *Journal of Geophysical Research*, *95*(B7), 10869–10,884. https://doi.org/10.1029/JB095iB07p10869

Irving, E., Park, J. K., & Mcglynn, J. C. (1972). Paleomagnetism of the Et-Then Group and Mackenzie diabase in the Great Slave Lake area. *Canadian Journal of Earth Sciences*, *9*(6), 744–755. https://doi.org/10.1139/e72-061

Kean, W. F., Williams, I., Chan, L., & Feeney, J. (1997). Magnetism of the Keweenawan age Chengwatana lava flows, northwest Wisconsin. *Geophysical Research Letters*, *24*(12), 1523–1526. https://doi.org/10.1029/97GL00993

Krogh, T. E., Corfu, F., Davis, D. W., Dunning, G. R., Heaman, L. M., Kamo, S. L., et al. (1987). Precise U-Pb isotopic ages of diabase dykes and mafic to ultramafic rocks using trace amounts of baddeleyite and zircon. In H. C. Halls & W. F. Fahrig (Eds.), *Mafic dyke swarms* (pp. 147–152). Geological Association of Canada, Special Paper 34.

Krumrei, T. V., Villa, I. M., Marks, M. A. W. W., & Markl, G. (2006). A $^{40}Ar/^{39}Ar$ and U/Pb isotopic study of the Ilímaussaq complex, South Greenland: Implications for the 40K decay constant and for the duration of magmatic activity in a peralkaline complex. *Chemical Geology*, *227*(3–4), 258–273. https://doi.org/10.1016/j.chemgeo.2005.10.004

Kulakov, E., Smirnov, A., & Diehl, J. (2013). Paleomagmatism of ~1.09 Ga Lake Shore Traps (Keweenaw Peninsula, Michigan): New results and implications. *Canadian Journal of Earth Sciences*, *50*(11), 1085–1096. https://doi.org/10.1139/cjes-2013-0003

Kulakov, E., Smirnov, A. V., & Diehl, J. F. (2014). Paleomagnetism of the ~1.1 Ga Coldwell complex (Ontario, Canada): Implications for Proterozoic geomagnetic field morphology and plate velocities. *Journal of Geophysical Research*, *119*, 8633–8654. https://doi.org/10.1002/2014JB011463

Larsen, L. M. (2006). Mesozoic to Palaeogene dyke swarms in West Greenland and their significance for the formation of the Labrador Sea and the Davis Strait. Danmarks og Grønlands Geologiske Undersøgelse Rapport 2006/34.

LeCheminant, A. N., & Heaman, L. M. (1989). Mackenzie igneous events, Canada: Middle Proterozoic hotspot magmatism associated with ocean opening. *Earth and Planetary Science Letters*, *96*(1–2), 38–48. https://doi.org/10.1016/0012-821X(89)90122-2

McFadden, P. L., & McElhinny, M. (1988). The combined analysis of remagnetisation circles and direct observation in palaeomagnetism. *Earth and Planetary Science Letters*, *87*(1-2), 161–172. https://doi.org/10.1016/0012-821X(88)90072-6

McFadden, P. L., & McElhinny, M. W. (1990). Classification of the reversal test in palaeomagnetism. *Geophysical Journal International*, *103*(3), 725–729. https://doi.org/10.1111/j.1365-246X.1990.tb05683.x

Meert, J. G., van der Voo, R., Powell, C. M., Li, Z.-X., McElhinny, M. W., Chen, Z., & Symons, D. T. A. (1993). A plate-tectonic speed limit? *Nature*, *363*(6426), 216–217. https://doi.org/10.1038/363216a0

Ojakangas, R. W., Morey, G. B., & Green, J. C. (2001). The Mesoproterozoic Midcontinent Rift System, Lake Superior Region, USA. *Sedimentary Geology*, *141–142*, 421–442.

Palmer, H. C., & Davis, D. W. (1987). Paleomagnetism and U-Pb geochronology of volcanic rocks from Michipicoten Island, Lake Superior, Canada: Precise calibration of the Keweenawan polar wander track. *Precambrian Research*, *37*(2), 157–171. https://doi.org/10.1016/0301-9268(87)90077-5

Palmer, H. C., Merz, B. A., & Hayatsu, A. (1977). The Sudbury dikes of the Grenville Front region: Paleornagnetism, petrochemistry, and K-Ar age studies. *Canadian Journal of Earth Sciences*, *14*(8), 1867–1887. https://doi.org/10.1139/e77-158

Park, J. K., & Roy, J. L. (1979). Further paleomagnetic results from the Seal Group igneous rocks, Quebec. *Canadian Journal of Earth Sciences*, *16*(4), 895–912. https://doi.org/10.1139/e79-078

Paslick, C. R., Halliday, A. N., Davies, G. R., Mezger, K., & Upton, B. G. J. (1993). Timing of Proterozoic magmatism in the Gardar Province, southern Greenland. *Geological Society of America Bulletin*, *105*(2), 272–278. https://doi.org/10.1130/0016-7606(1993)105<0272:TOPMIT>2.3.CO;2

Pearce, J. A. (2008). Geochemical fingerprinting of oceanic basalts with applications to ophiolite classification and the search for Archean oceanic crust. *Lithos*, *100*(1–4), 14–48. https://doi.org/10.1016/j.lithos.2007.06.016

Pesonen, L. J. (1979). Paleomagnetism of late Precambrian Keweenawan igneous and baked contact rocks from Thunder Bay District, northern Lake Superior. *The Geological Society of Finland, Bulletin*, *51*(1-2), 27–44. https://doi.org/10.17741/bgsf/51.1-2.004

Pesonen, L. J., & Nevanlinna, H. (1981). Late Precambrian Keweenawan asymmetric reversals. *Nature*, *294*(5840), 436–439. https://doi.org/10.1038/294436a0

Piper, J. D. A. (1977). Palaeomagnetism of the giant dykes of Tugtutoq and Narssaq gabbro, Gardar Igneous Province, South Greenland. *Bulletin of the Geological Society of Denmark*, *26*, 85–94.

Piper, J. D. A. (1992). The palaeomagnetism of major (middle proterozoic) igneous complexes, South Greenland and the Gardar apparent polar wander track. *Precambrian Research*, *54*(2–4), 153–172. https://doi.org/10.1016/0301-9268(92)90068-Y

Piper, J. D. A. (2010). Palaeopangaea in Meso-Neoproterozoic times: The palaeomagnetic evidence and implications to continental integrity, supercontinent form and Eocambrian break-up. *Journal of Geodynamics*, *50*(3–4), 191–223. https://doi.org/10.1016/j.jog.2010.04.004

Piper, J. D. A., Thomas, D. N., Share, S., & Rui, Z. Q. (1999). The palaeomagnetism of (Mesoproterozoic) Eriksfjord group red beds, South Greenland: Multiphase remagnetization during the Gardar and Grenville episodes. *Geophysical Journal International*, *136*(3), 739–756. https://doi.org/10.1046/j.1365-246x.1999.00756.x

Pisarevsky, S. A., Elming, S. Å., Pesonen, L. J., & Li, Z. X. (2014). Mesoproterozoic paleogeography: Supercontinent and beyond. *Precambrian Research*, *244*(1), 207–225. https://doi.org/10.1016/j.precamres.2013.05.014

Platt, R., Mitchell, R., & Holm, P. (1983). Marathon dikes: Rb-Sr and K-Ar geochronology of ultrabasic lamprophyres from the vicinity of McKellar Harbour, northwestern Ontario, Canada. *Canadian Journal of Earth Sciences*, *20*(6), 961–967. Retrieved from https://doi.org/10.1139/e83-085.

Platt, R. G., & Mitchell, R. H. (1979). The Marathon dikes. I: Zirconium-rich titanian garnets and manganoan magnesian ulvöspinel-magnetite spinels. *American Mineralogist*, *64*, 546–550.

Platt, R. G., & Mitchell, R. H. (1982). The Marathon dikes: Ultrabasic lamprophyres from the vicinity of McKellar harbour, N.W. Ontario. *American Mineralogist*, *67*(9–10), 907–916.

Queen, M., Hanes, J. A., Archibald, D. A., Heaman, L. M., Hanes, J. A., Archibald, D. A., & Farrar, E. (1996). $^{40}Ar/^{39}Ar$ phlogopite and U-Pb perovskite dating of lamprophyre dykes from the eastern Lake Superior region: Evidence for a 1.1 4 Ga magmatic precursor to Midcontinent Rift volcanism. *Canadian Journal of Earth Sciences*, *33*(6), 958–965. https://doi.org/10.1139/e96-072

Robertson, W. A., & Fahrig, W. F. (1971). The Great Logan Paleomagnetic Loop—The polar wandering path from Canadian shield rocks during the Neohelikian era. *Canadian Journal of Earth Sciences*, *8*(11), 1541–1555.

Rock, N. M. S. (1986). The nature and origin of ultramafic lamprophyres: Alnoites and allied rocks. *Journal of Petrology*, *27*(1), 155–196. https://doi.org/10.1093/petrology/27.1.155

Rock, N. M. S. (1987). The nature and origin of lamprophyres: An overview. In J. G. Fitton & B. G. J. Upton (Eds.). *Alkaline Igneous Rocks. Geological Society, London, Special Publications*, *30*, 191–226. https://doi.org/10.1144/GSL.SP.1987.030.01.09

Roest, W. R., & Srivastava, S. P. (1989). Sea-floor spreading in the Labrador Sea: A new reconstruction. *Geology*, *17*(11), 1000–1003. https://doi.org/10.1130/0091-7613(1989)017<1000:SFSITL>2.3.CO;2

Roy, J. L., & Robertson, W. A. (1978). Paleomagnetism of the Jacobsville Formation and the apparent polar path for the interval −1100 to −670 m.y. or North America. *Journal of Geophysical Research*, *83*(B3), 1289. https://doi.org/10.1029/JB083iB03p01289

Rukhlov, A. S., & Bell, K. (2010). Geochronology of carbonatites from the Canadian and Baltic Shields, and the Canadian Cordillera: Clues to mantle evolution. *Mineralogy and Petrology*, *98*(1-4), 11–54. https://doi.org/10.1007/s00710-009-0054-5

Sage, R. P. (1991a). Alkalic rock, carbonatite and kimberlite complexes of Ontario, Superior Province. In P. C. Thurston, H. R. Williams, R. H. Sutcliffe, & G. M. Stott (Eds.), *Geology of Ontario, Part 1* (Vol. 1, pp. 683–709). Ontario: Ontario Geological Survey, Special Volume.

Sage, R. P. (1991b). Precambrian geology: Slate Islands. Ontario Geological Survey Report (Vol. 264).

Sage, R. P. (2000). The"Sandor" diamond Occurence, Michipicoten Greenstone Belt, Wawa, Ontario: A preliminary study. Ontario Geological Survey Open File Report, 6106, 49.

Schmidt, P. W. (1993). Palaeomagnetic cleaning strategies. *Physics of the Earth and Planetary Interiors*, *76*(1–2), 169–178. https://doi.org/10.1016/0031-9201(93)90066-I

Smirnov, A. V., Tarduno, J. A., & Evans, D. A. D. (2011). Evolving core conditions ca. 2 billion years ago detected by paleosecular variation. *Physics of the Earth and Planetary Interiors*, *187*(3–4), 225–231. https://doi.org/10.1016/j.pepi.2011.05.003

Steinberger, B., & Torsvik, T. H. (2008). Absolute plate motions and true polar wander in the absence of hotspot tracks. *Nature*, *452*(7187), 620–623. https://doi.org/10.1038/nature06824

Stott, G. M., & Josey, S. D. (2009). Post-Archean mafic (diabase) dikes and other intrusions of northwestern Ontario, north of latitude 49°30. Ontario Geological Survey, Miscellaneous Release - Data, (241).

Sun, S. -s., & McDonough, W. F. (1989). Chemical and isotopic systematics of oceanic basalts: Implications for mantle composition and processes. *Geological Society, London, Special Publications*, *42*(1), 313–345. https://doi.org/10.1144/GSL.SP.1989.042.01.19

Swanson-Hysell, N. L., Maloof, A. C., Weiss, B. P., & Evans, D. a. D. (2009). No asymmetry in geomagnetic reversals recorded by 1.1-billion-year-old Keweenawan basalts. *Nature Geoscience*, *3*(5), 369–717. https://doi.org/10.1038/ngeo850

Swanson-Hysell, N. L., Vaughan, A. A., Mustain, M. R., & Asp, K. E. (2014). Confirmation of progressive plate motion during the Midcontinent Rift's early magmatic stage from the Osler Volcanic Group, Ontario, Canada. *Geochemistry, Geophysics, Geosystems*, *15*, 2039–2047. https://doi.org/10.1002/2013GC005180

Symons, D. T. a. (1992). Paleomagnetism of the Keweenawan Chipman Lake and Seabrook Lake carbonatite complexes, Ontario. *Canadian Journal of Earth Sciences*, *29*(6), 1215–1223. https://doi.org/10.1139/e92-097

Tappe, S., Foley, S. F., Kjarsgaard, B. A., Romer, R. L., Heaman, L. M., Stracke, A., & Jenner, G. A. (2008). Between carbonatite and lamproite-diamondiferous Torngat ultramafic lamprophyres formed by carbonate-fluxed melting of cratonic MARID-type metasomes. *Geochimica et Cosmochimica Acta*, *72*(13), 3258–3286. https://doi.org/10.1016/j.gca.2008.03.008

Tauxe, L., & Kodama, K. (2009). Paleosecular variation models for ancient times: Clues from Keweenawan lava flows. *Physics of the Earth and Planetary Interiors*, *177*(1–2), 31–45. https://doi.org/10.1016/j.pepi.2009.07.006

Tauxe, L., Mullender, T. a. ., & Pick, T. (1996). Potbellies, wasp-waists, and superparamegnetism in magnetic hysteresis. *Journal of Geophysical Research* https://doi.org/10.1029/95JB03041, *101*(B1), 571–583.

Upton, B. G. J. (2013). Tectono-magmatic evolution of the younger Gardar southern rift, South Greenland. *Geological Survey of Denmark & Greenland Bulletin*, *29*.

Upton, B. G. J., Emeleus, C. H., Heaman, L. M., Goodenough, K. M., & Finch, a. a. (2003). Magmatism of the mid-Proterozoic Gardar Province, South Greenland: Chronology, petrogenesis and geological setting. *Lithos*, *68*(1-2), 43–65. https://doi.org/10.1016/S0024-4937(03)00030-6.

Van der Voo, R. (1990). The reliability of paleomagnetic data. *Tectonophysics*, *184*(1), 1–9. https://doi.org/10.1016/0040-1951(90)90116-P

Van Schmus, W. R., & Hinze, W. J. (1985). The Midcontinent Rift System. *Annual Review of Earth and Planetary Sciences*, *13*(1), 345–383. https://doi.org/10.1146/annurev.ea.13.050185.002021

Vandamme, D. (1994). A new method to determine paleosecular variation. *Physics of the Earth and Planetary Interiors*, *85*(1–2), 131. https://doi.org/10.1016/0031-9201(94)90012-4–142.

Veikkolainen, T. H., Biggin, A. J., Pesonen, L. J., Evans, D. A., & Jarboe, N. A. (2017). Advancing Precambrian palaeomagnetism with the PALEOMAGIA and PINT(QPI) databases. *Scientific Data*, *4*(170068), 170,068. https://doi.org/10.1038/sdata.2017.68

Verwey, E. J. W. (1939). Electronic conduction of magnetite (Fe_3O_4) and its transition point at low temperatures. *Nature*, *144*(3642), 327–328. https://doi.org/10.1038/144327b0.

Walker, J. W. R. (1967). Geology of the Jackfish-Middleton Area, District of Thunder Bay. Ontario Department of Mines, Geological Report, 50, 41 p.

Wu, F. Y., Yang, Y. H., Marks, M. A. W., Liu, Z. C., Zhou, Q., Ge, W. C., et al. (2010). In situ U-Pb, Sr, Nd and Hf isotopic analysis of eudialyte by LA-(MC)-ICP-MS. *Chemical Geology*, *273*(1–2), 8–34. https://doi.org/10.1016/j.chemgeo.2010.02.007